

Relative Covariance

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September 9, 2026

***Note:** Parts of this document may be AI-written. My general workflow is to write sections of the article myself, laying out the general points I want to hit on and going into detail where relevant. I then converse with an LLM to identify errors in the logic or math and opportunities to use clearer or more generalizable notation. I then have the LLM make the edits to reflect our proposed changes, including adding more detailed derivations.*

This article describes how the phenotypic covariance between two individuals i and j can be modeled by limited information we have about the pair. I derive formulas from the starting trait generative model and only introduce assumptions at the point the derivation requires it. I start with a simple trait architecture and iteratively add more complexity.

1 Additive genetics under random mating with independent environment

We begin with the simplest architecture: a trait determined by an additive genetic component and an independent environmental component, in a randomly mating population.

1.1 The phenotype model

For an individual i we write the (standardized) phenotype as

$$y_i = g_i + e_i, \tag{1}$$

where g_i is the genetic value and e_i the environmental deviation. We take y to be standardized across the population, $E[y] = 0$ and $\text{Var}[y] = 1$, and both components to be mean-centered, $E[g] = E[e] = 0$, with variances $\text{Var}[g] = V_A$ and $\text{Var}[e] = V_E$.

We then introduce the following assumptions.

[Additivity] Additive genetic value

The genetic value is a fixed additive function of the individual's own genotype,

$$g_i = \sum_{k=1}^M \beta_k x_{ik},$$

where β_k is the per-standardized-allele effect of variant k , x_{ik} is the standardized genotype ($E[x_{ik}] = 0$, $\text{Var}[x_{ik}] = 1$), and M is the number of variants in the genome. Non-causal variants are included with $\beta_k = 0$.

Commonly bundled into: the additive model; no dominance, no gene–gene interactions (GxG, epistasis), and no gene–environment interactions (GxE) — each would make the genetic value depend on more than the individual’s own genotype in an additive way.

[GE-indep] Gene–environment independence

An individual’s environmental component is independent of their own genetic component:

$$g_i \perp e_i$$

Commonly bundled into: no gene–environment correlation or interaction, within an individual.

[No-IGE] No indirect genetic effects

One individual’s genetic component is independent of another individual’s environmental component:

$$g_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no indirect (e.g. parental) genetic effects — one individual’s genotype does not shape another’s environment.

[Env-indep] Independent environments

The environmental component is independent across individuals, including relatives:

$$e_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no shared/common environment (C); no environmental confounder shared by the pair (e.g. population stratification).

Because $g_i \perp e_i$ ([GE-indep]), $\text{Cov}[g_i, e_i] = 0$ and the phenotypic variance decomposes as $\text{Var}[y] = V_A + V_E$. Narrow-sense heritability is typically defined as:

$$h^2 = \frac{V_A}{\text{Var}[y]}$$

Since $\text{Var}[y] = 1$, $h^2 = V_A$ under this model.

1.2 Phenotypic covariance is genotypic covariance

For a single fixed pair, y_i and y_j are two fixed values, so “their covariance” only has meaning as an average over an ensemble of pairs. The ensemble we use is the set of all pairs that share a given piece of *relatedness information* $\mathcal{I}_{ij}$ about the pair, and the object of interest is the conditional cross-product

$$C_{ij}(\mathcal{I}) := \mathbb{E}[y_i y_j \mid \mathcal{I}_{ij}]. \quad (2)$$

If $\mathbb{E}[y_i \mid \mathcal{I}] = 0$, then $C_{ij}(\mathcal{I}) = \text{Cov}[y_i, y_j \mid \mathcal{I}]$. For most information $\mathcal{I}$, like the realized relatedness or degree between two relatives, this equivalence is applicable. An exception is when the information we are modelling contains individuals’ actual genotype information, for which $\mathbb{E}[y_i \mid \text{genotypes}] = g_i \neq 0$, therefore the expected cross-product of two individuals is not equivalent to their covariance given the fixed information.

Using (1) together with [No-IGE] and [Env-indep], every cross term involving e vanishes in expectation (each factors through a mean-zero environmental term), so the phenotypic cross-product reduces to the genetic one:

$$C_{ij}(\mathcal{I}) = \mathbb{E}[(g_i + e_i)(g_j + e_j) \mid \mathcal{I}] = \mathbb{E}[g_i g_j \mid \mathcal{I}]. \quad (3)$$

The rest of the article evaluates (3) at four levels of resolution for $\mathcal{I}$.

1.3 Level 0 — the exact genetic cross-product

At the finest resolution we condition on the realized genotypes of i and j at the causal variants, for which we know the true causal effect sizes. Then g_i and g_j are fixed, and by [Additivity],

$$C_{ij} = \mathbb{E}[y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j, \boldsymbol{\beta}] = g_i g_j = \sum_k \sum_{k'} \beta_k \beta_{k'} x_{ik} x_{jk'} \quad (4)$$

Under our model and assumptions so far, this returns the true expected cross-product of two individuals’ phenotypes. The genetic cross-product between two individuals is, fundamentally, an effect-size-weighted function of the genotypes they actually carry at causal sites. Notably, this formula does not require any knowledge of the relatedness of the two individuals.

1.4 Level 1 — genome-wide allelic sharing

If we do not know the causal effect sizes, then we cannot apply (4). However, we can assume that genetic effects’ cross-products are independent of individuals’ allelic sharing patterns.

[Random-effects] Random effects, exchangeable across variants

Treating the effects as random, they are mean-zero and uncorrelated across variants, the distribution of their cross-products $\{\beta_k\beta_{k'}\}$ is independent of the genotypes $\mathbf{X}$, and every variant carries the same second moment:

$$\mathbb{E}[\beta_k] = 0, \quad \mathbb{E}[\beta_k\beta_{k'}] = 0 \quad (k \neq k'), \quad \mathbb{E}[\beta_k\beta_{k'} \mid \mathbf{X}] = \mathbb{E}[\beta_k\beta_{k'}], \quad \mathbb{E}[\beta_k^2] = \mathbb{E}[\beta_{k'}^2].$$

The last condition is a statement about the *prior* we place on effects, not about their realized values: it says we have no information distinguishing one variant from another, so we weight them symmetrically.

Commonly bundled into: the infinitesimal / random-effects model.

Taking the expectation of (4) over the effects, and using the independence of the cross-products from the genotypes ([Random-effects]) to replace $\mathbb{E}[\beta_k\beta_{k'} \mid \mathbf{X}]$ by $\mathbb{E}[\beta_k\beta_{k'}]$:

$$\mathbb{E}[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] = \sum_k \sum_{k'} \mathbb{E}[\beta_k\beta_{k'}] x_{ik} x_{jk'}.$$

Because the effects are mean-zero and uncorrelated across variants, $\mathbb{E}[\beta_k\beta_{k'}] = 0$ whenever $k \neq k'$; the entire off-diagonal ($k \neq k'$) collapses regardless of any correlation between the genotypes. What remains is the diagonal, weighted by each variant's effect variance:

$$\mathbb{E}[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] = \sum_k \mathbb{E}[\beta_k^2] x_{ik} x_{jk}.$$

The additive variance is the total of these: since $\text{Var}[x_{ik}] = 1$ and the off-diagonal effect terms vanish, $V_A = \text{Var}[g_i] = \sum_k \mathbb{E}[\beta_k^2]$, and thus $\mathbb{E}[\beta_k^2] = V_A/M$. Effectively, the per-variant effect weighting of (4) has been replaced by a uniform weighting across all variants. Therefore

$$C_{ij}(\mathbf{x}_i, \mathbf{x}_j) = V_A \cdot \frac{1}{M} \sum_k x_{ik} x_{jk}. \quad (5)$$

The quantity multiplying V_A is the genome-wide genotype cross-product (i.e. allelic sharing), which we'll define by:

$$A_{ij} = \frac{1}{M} \sum_k x_{ik} x_{jk}$$

This can be thought of as an identity-by-state (IBS)-based measure of “relatedness”, and is exactly the (i, j) entry of what is typically computed as a Genetic Relatedness Matrix (GRM), $\mathbf{A}$. Therefore, we can further simplify:

$$\boxed{C_{ij}(A_{ij}) = V_A A_{ij}} \quad (6)$$

1.5 Level 2 — realized relatedness

The problem with calculating A_{ij} is that we may not have genotyped every causal variant that contributes to a trait. As long as some of the causal variants are not in perfect

correlation with a genotyped variant, then the covariance of two individuals' phenotypes will not fully take into account the contributions of the ungenotyped causal variants. However, when individuals are related, their IBS-based relatedness from genotyped variants is informative of the IBS-based relatedness of ungenotyped variants. Therefore, the former can be used to capture the contributions of the latter.

When individuals inherit a genomic segment from a common ancestor, then all of the variants within that genomic segment will be identical-by-state, as long as we assume that de-novo mutations have not caused the inherited alleles to be different. These genomic segments and their variants are said to be identical-by-descent (IBD).

[No-mutation] IBD implies IBS

Alleles that are identical-by-descent are also identical-by-state: descent is transmitted without mutation.

Commonly bundled into: the infinite-sites / no-recurrent-mutation assumption.

When two individuals' alleles are IBD, then they are always IBS; when they are not IBD, they can still be IBS. Therefore, we can decompose the IBS-based relatedness A_{ij} into a sum over IBD allele pairs and a sum over non-IBD allele pairs. Write each standardized genotype as the sum of its two per-allele (maternal and paternal) standardized values, $x_{ik} = z_{ik}^{(m)} + z_{ik}^{(p)}$. Since $\text{Var}[x_{ik}] = 1$ and the two alleles are independent ([No-inbreeding]), each per-allele value has $E[z] = 0$ and $\text{Var}[z] = \frac{1}{2}$. The genotype cross-product then expands over the four ordered allele pairs, and A_{ij} splits by IBD status:

$$\begin{aligned} A_{ij} &= \frac{1}{M} \sum_k x_{ik} x_{jk} = \frac{1}{M} \sum_k \sum_{u,v \in \{m,p\}} z_{ik}^{(u)} z_{jk}^{(v)} \\ &= \underbrace{\frac{1}{M} \sum_{\text{IBD pairs}} z_{ik}^{(u)} z_{jk}^{(v)}}_{\text{IBD}} + \underbrace{\frac{1}{M} \sum_{\text{non-IBD pairs}} z_{ik}^{(u)} z_{jk}^{(v)}}_{\text{non-IBD}}. \end{aligned} \tag{7}$$

For an IBD allele pair, the two per-allele values are copies of the same ancestral allele and, by [No-mutation], of the same state; they are therefore equal, so their expected cross-product is $E[(z_{ik}^{(u)})^2] = \text{Var}[z_{ik}^{(u)}] = \frac{1}{2}$. For non-IBD allele pairs, we have to make an assumption on how they are distributed.

[Non-IBD-indep] Independence of non-IBD alleles

Any two alleles that are *not* IBD are independent draws from the population allele distribution.

Commonly bundled into: between individuals: random mating, no population structure, and no assortative mating.

[No-inbreeding] No inbreeding

Within an individual, the two homologous alleles are not IBD with each other, and hence (by [Non-IBD-indep]) independent. This results in each variant's genotype variance being $2p_k(1 - p_k)$ — so the standardized x_{ik} has variance 1 — and it keeps the number of alleles a pair shares IBD at a variant in $\{0, 1, 2\}$.

Commonly bundled into: a randomly mating, outbred population (Hardy–Weinberg within individuals).

By the definition of independent draws, the expected cross-product of a non-IBD allele pair is therefore $E[z_{ik}^{(u)}]E[z_{jk}^{(v)}] = 0$. Letting $\text{IBD}_{ij}(k) \in \{0, 1, 2\}$ denote the number of alleles the pair shares IBD at variant k , exactly $\text{IBD}_{ij}(k)$ of the four ordered allele pairs are IBD — each contributing $\frac{1}{2}$ — and the rest contribute 0, so

$$E[x_{ik}x_{jk} \mid \text{IBD}_{ij}(k)] = \frac{1}{2}\text{IBD}_{ij}(k).$$

A variant thus contributes expected cross-product 0, $\frac{1}{2}$, or 1 according to whether the pair shares 0, 1, or 2 alleles IBD there. We define the pair's *realized relatedness* π_{ij} as the average IBD state across variants, halved — equivalently, the fraction of the pair's genome that is IBD:

$$\pi_{ij} := \frac{1}{2M} \sum_k \text{IBD}_{ij}(k). \quad (8)$$

Conditioning on the pair's realized IBD states and using the per-variant identity $E[x_{ik}x_{jk} \mid \text{IBD}_{ij}(k)] = \frac{1}{2}\text{IBD}_{ij}(k)$, linearity of expectation gives the expected allelic sharing:

$$E[A_{ij} \mid \{\text{IBD}_{ij}(k)\}] = \frac{1}{M} \sum_k \frac{1}{2} \text{IBD}_{ij}(k) = \frac{1}{2M} \sum_k \text{IBD}_{ij}(k) = \pi_{ij}.$$

This depends on the IBD states only through π_{ij} , so we may condition on π_{ij} alone. Crucially, it required no assumption that the per-variant IBD states are independent — and indeed they are not: an IBD state changes only at a cross-over within a co-inherited segment, so physically linked variants have strongly correlated IBD states. Linearity of expectation is insensitive to this correlation.

Combining with (6), the pair's expected phenotypic cross-product is their realized relatedness scaled by the additive variance:

$$\boxed{E[y_i y_j \mid \pi_{ij}] = V_A E[A_{ij} \mid \pi_{ij}] = V_A \pi_{ij}}. \quad (9)$$

1.6 Level 3 — genetic degree

In cases where we cannot measure π_{ij} based on per-variant IBD states, we can still estimate it if we know the genetic degree d_{ij} between the individuals.

[Genetic-degree] Degree determines expected relatedness

The genetic degree d_{ij} fixes the distribution of IBD sharing through Mendelian segregation, so that the expected relatedness given the degree is a single number $\bar{\pi}_{d_{ij}}$:

$$E[\pi_{ij} \mid d_{ij}] = \bar{\pi}_{d_{ij}}.$$

Furthermore, under the outbreeding assumption ([No-inbreeding]) — applied here not only to i and j but to the entire pedigree connecting them, so that it contains no inbreeding loops — $\bar{\pi}_{d_{ij}} = 2^{-d_{ij}}$, which is not fully derived here. We can then obtain the expected phenotypic covariance given the degree d_{ij} :

$$E[y_i y_j \mid d_{ij}] = E[E[y_i y_j \mid \pi_{ij}] \mid d_{ij}] = V_A E[\pi_{ij} \mid d_{ij}] = V_A \bar{\pi}_{d_{ij}}. \quad (10)$$

1.7 Summary

Each level conditions on coarser information than the one above and is its conditional expectation; the rightmost column lists the assumption(s) newly required to descend one step.

Explicitly, each level is the conditional expectation of the finer level above it, averaging out one additional layer of information:

$$\begin{aligned} E[y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j] &= E[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] = V_A A_{ij}, \\ E[y_i y_j \mid \pi_{ij}] &= E[V_A A_{ij} \mid \pi_{ij}] = V_A \pi_{ij}, \\ E[y_i y_j \mid d_{ij}] &= E[V_A \pi_{ij} \mid d_{ij}] = V_A \bar{\pi}_{d_{ij}}, \end{aligned}$$

starting from the Level-0 value $E[y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j, \boldsymbol{\beta}] = g_i g_j$. Equivalently, the ladder telescopes into a single nested expectation,

$$E[y_i y_j \mid d_{ij}] = E\left[E\left[E[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] \mid \pi_{ij}\right] \mid d_{ij}\right] = V_A \bar{\pi}_{d_{ij}}.$$

This is valid down the generative chain $d_{ij} \rightarrow \pi_{ij} \rightarrow (\mathbf{x}_i, \mathbf{x}_j) \rightarrow y$: conditioned on the finer variable, the phenotype does not depend further on the coarser one, so the tower property applies at each step.

Level	Conditioned on	$E[y_i y_j \mid \cdot]$	New assumptions
0	causal genotypes	$g_i g_j$	[Additivity] (given [No-IGE], [Env-indep])
1	realized sharing A_{ij}	$V_A A_{ij}$	[Random-effects]
2	realized relatedness π_{ij}	$V_A \pi_{ij}$	[No-mutation], [Non-IBD-indep], [No-inbreeding]
3	genetic degree d_{ij}	$V_A \bar{\pi}_{d_{ij}}$	[Genetic-degree]

The property that makes this a ladder is worth naming, because the next architecture breaks it. At each step, the finer variable *screens off* the coarser one: once we condition on π_{ij} , knowing d_{ij} as well would tell us nothing more about $y_i y_j$, and likewise for the other steps. Screening is what licenses the tower property, and it is the reason the four information sets can be arranged in a single descending chain rather than a more complicated structure.

2 Additive genetics under assortative mating

We will now do away with the assumption of random mating (particularly implied in [\[Non-IBD-indep\]](#)) and instead provide a model for assortative mating that will let us quantify its impact on phenotypic covariance.

The phenotype model remains the same: we assume an additive genetic architecture ([\[Additivity\]](#)) with an independent environmental component ([\[GE-indep\]](#)). As a consequence of the assortative mating model we adopt below, covariance between the environmental and genetic components between some individuals will arise, breaking the weakest forms of the assumptions [\[No-IGE\]](#) and [\[Env-indep\]](#). We also still assume no new mutations ([\[No-mutation\]](#)) and no inbreeding ([\[No-inbreeding\]](#)).

2.1 The mating model

We assume a model of assortative mating based on mates' similarity in a sorting factor, which is also our focal trait.

[\[Primary-AM\]](#) Primary Assortative Mating

Mating is non-random such that two mates' phenotypes are correlated for some trait at the time of mating:

$$\text{Corr}[y_m, y_p] = \rho_y,$$

where the subscripts m and p index the maternal and paternal member of a mated pair. Matching is on the phenotype *only*: any covariance between the mates' trait components is a downstream consequence of this phenotypic matching, and is not specified separately. Following the convention that a subscript names the component being correlated, we write ρ_g for the induced correlation between mates' genetic values: $\text{Corr}[g_m, g_p] = \rho_g^{(t)}$, where the superscript records that this induced correlation changes from one generation to the next, unlike ρ_y , which is held fixed across generations.

Commonly bundled into: primary phenotypic assortment, as opposed to social homogamy (matching on a shared environment) or genetic homogamy (matching on genotype directly); it also excludes convergence, where mates grow more similar after mating. ρ_y is often written μ or r in the literature.

2.2 Dynamics and equilibrium

To make modelling the dynamics of populations easier, we will make the following assumption:

[Discrete-generations] Discrete, non-overlapping generations

The population is partitioned into generations indexed by t , and every individual mates within their own generation.

Commonly bundled into: no overlapping generations and no age structure; in particular no mating between individuals of different generations, and no individual contributing offspring to more than one generation.

When starting from a randomly mating population, one generation of assortative mating causes the immediate offspring generation to exhibit different phenotypic and genetic properties on expectation. Luckily, if the parameters of assortative mating remain fixed, an equilibrium is reached such that stable properties of the population can be studied. We will first explore how assortative mating affects statistical properties of the population during the first few generations, then we will solve for the parameter values that lead to an equilibrium state.

2.2.1 Path Analysis

First, we'll introduce path analysis, a very helpful tool for modelling the covariance between variables in a causal system, such as genetic inheritance and assortative mating. Path analysis was introduced by Sewall Wright [1], as a way of decomposing an observed correlation between variables into the contributions of a hypothesized causal structure, and set out systematically thirteen years later as the method of path coefficients [2]. Wright applied it to mating systems in the same period [3, 4]; together with Fisher's treatment of the same problem [5], that work is the origin of the classical results on how assortment changes the correlation between relatives — the results this section rederives. The explanations below are adapted from the Supplementary Note of Sunde et al. [6], which introduced the co-path edge type (described below). Unlike Sunde et al. [6], we do not assume that the variables are standardized, so we will keep track of variances explicitly.

A *path diagram* is a graphical representation of a generative model. Each variable is a node — drawn as a rectangle if it is observed and an ellipse if it is latent — and the nodes are joined by three kinds of edges:

- A **directed** edge $a \rightarrow b$ labelled with a coefficient c says that a is a cause of b , contributing ca to b 's structural equation. Collecting the arrows that point *into* a node recovers that node's equation, so the diagram and the model equations carry the same content.
- A **bidirected** edge $a \leftrightarrow b$ carries covariance between the *disturbances* of a and b . A variable's disturbance is what remains of it once its causes in the diagram

are accounted for: if the arrows into b give $b = c_1 a_1 + \dots + c_k a_k + \epsilon_b$, then ϵ_b is b 's disturbance, and for a variable with no incoming arrows the disturbance is the variable itself. The self-loop $a \leftrightarrow a$ is the case $b = a$: it is $\text{Var}[\epsilon_a]$, the part of $\text{Var}[a]$ not supplied by a 's causes, so a variable with no incoming arrows carries its entire variance on such a loop.

- A **co-path** $a \text{ --- } b$, drawn as a plain line with no arrowheads, is covariance attributable to *matching*: a and b came to be associated because they were paired on their values, not because they share a cause.

The covariance between *any* two variables in the diagram, latent or observed, can be calculated quickly by following Wright's tracing rules. Enumerate every *valid chain* joining the two variables; their covariance is the sum over chains of the product of the coefficients each chain passes through. A valid chain between a and b has the shape

$$a \leftarrow \dots \leftarrow u \leftrightarrow v \rightarrow \dots \rightarrow b :$$

trace *backward* along directed edges from a to some ancestor u , cross *exactly one* bidirected edge $u \leftrightarrow v$ (possibly a self-loop, $u = v$), then trace *forward* along directed edges from v down to b . A chain turns around exactly once, and it turns around on a bidirected edge, because that is the only place variance enters the system.

Co-paths extend this. A chain may pass *through* a co-path and keep going: a co-path joins two valid chains of the kind above into a longer one,

$$\underbrace{a \leftarrow \dots \leftrightarrow \dots \rightarrow y_m}_{\text{valid chain}} \text{ --- } \underbrace{y_p \leftarrow \dots \leftrightarrow \dots \rightarrow b}_{\text{valid chain}} .$$

Two further rules apply [6]: a chain can neither start nor end with a co-path, and a chain cannot use two co-paths belonging to the same mating process. It *may* cross co-paths of different mating processes.

A bidirected edge is a statement about disturbances, so a chain terminates on it: a disturbance has no causes in the diagram, so there is nothing to keep tracing back through, and the chain cannot reach past y_m to the things that caused y_m . A co-path is a statement about the matched variables themselves, so a chain continues backward through it, which encodes the fact that matching two people on their phenotypes induces correlation between *all the causes* of those phenotypes — their genetic values, and every individual variant behind them. This is the mathematical basis behind the **[Primary-AM]** assumption.

The quantity multiplied along a chain when it crosses a co-path is not the correlation ρ_y itself, but

$$\mu^{(t)} = \frac{\rho_y}{\sqrt{\text{Var}[y_m^{(t)}] \text{Var}[y_p^{(t)}]}} = \frac{\rho_y}{V_P^{(t)}}, \quad (11)$$

because a co-path with coefficient μ contributes $\text{Cov}[a, b] = \mu \text{Var}[a] \text{Var}[b]$ [6]. The two coincide only when the matched variables have unit variance. Note the use of a superscript to denote the generation a quantity belongs to, since $V_P^{(t)}$ changes from one generation to the next under assortative mating (described below). A value of μ carried unchanged from one generation to the next is therefore wrong, which is the practical reason we label a co-path with its correlation in brackets, as $[\rho_y]$, and leave $\mu^{(t)}$ to be derived from the diagram's own variances.

Tracing chains by hand is instructive for a single pair and hopeless for a pedigree, since the number of chains grows multiplicatively with depth. The diagrams and covariances in this section were produced with **pathMgr**, a small package written alongside this writeup: it takes a model specification, returns the covariance between any two of its variables as an algebraic expression, and emits the path diagram to go with it. It computes each covariance in two independent ways — by enumerating chains exactly as above, and in closed form from the matrix identity $\mathbf{\Sigma} = \mathbf{F}(\mathbf{I} - \mathbf{A})^{-1}\mathbf{S}(\mathbf{I} - \mathbf{A})^{-\top}\mathbf{F}^\top$ — and their agreement is the correctness check.

That identity is the diagram written as matrices, one row and column per variable. Collecting all the variables into a vector $\mathbf{v}$, the matrix $\mathbf{A}$ holds the **directed** edges — A_{ij} is the coefficient on the arrow from j to i — so the whole system of structural equations is $\mathbf{v} = \mathbf{A}\mathbf{v} + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon}$ is the vector of disturbances. The symmetric matrix $\mathbf{S}$ holds the **bidirected** edges: S_{ij} is the value of $i \leftrightarrow j$, and its diagonal S_{ii} is the self-loop, the variance of i 's disturbance. $\mathbf{F}$ merely selects rows, keeping the variables we want reported and discarding the rest.

The inverse is what does the tracing. In an acyclic model $\mathbf{A}$ is nilpotent, so $(\mathbf{I} - \mathbf{A})^{-1} = \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \dots$, and the (i, u) entry of that sum is the total over all directed paths from u to i of the product of their coefficients — one term per path, $\mathbf{A}^n$ contributing the paths of length n . Sandwiching $\mathbf{S}$ between it and its transpose therefore gives, for latent and observed variables alike,

$$\Sigma_{ij} = \sum_u \sum_v \underbrace{[(\mathbf{I} - \mathbf{A})^{-1}]_{iu}}_{i \leftarrow \dots \leftarrow u} \underbrace{S_{uv}}_{u \leftrightarrow v} \underbrace{[(\mathbf{I} - \mathbf{A})^{-1}]_{jv}}_{v \rightarrow \dots \rightarrow j},$$

which is Wright's rule term for term: a backward leg, exactly one bidirected edge, a forward leg, summed over every chain, indexed by the $u \leftrightarrow v$ it turns around on. The two methods are the same statement — one enumerated, one summed in closed form — which is why their agreement tests the implementation rather than the mathematics. Co-paths are the one thing that sits outside this identity and contributes further terms; there, the chain rules given above are the definition.

2.2.2 Consequences of assortative mating on first generation of mates

We assume a population has been randomly mating as described by assumption **[Non-IBD-indep]** up to some generation $t = 0$, which can also be referred to as the base or randomly-mating population. The parameters defined in the previous section, such as the additive variance V_A and heritability h^2 , are measured in this base population and denoted by a (0), e.g. $V_A^{(0)}$ and $h_{(0)}^2$, since these terms do not remain fixed across generations under assortative mating. Note that V_E remains fixed.

Building the model from haploid alleles

To properly model the effects of assortative mating, it pays off to think in terms of individual haploid alleles, rather than diploid genotypes. We previously defined $z_{ik}^{(m)}$ and $z_{ik}^{(p)}$ as the standardized values of the maternally- and paternally-inherited alleles at

variant k for individual i , and $x_{ik} = z_{ik}^{(m)} + z_{ik}^{(p)}$ as the standardized diploid genotype. Each haploid allele $a^{(m)}$ and $a^{(p)}$ can take on two values, 0 or 1, and therefore each has a variance of $p_k(1 - p_k)$, where p_k is the allele frequency at variant k . Typically, diploid genotypes are standardized by their expected variance when the two alleles are independent, which is $2p_k(1 - p_k)$, resulting in a standardized variance of 1, as used in the previous section. Under this standardization, each haploid allele has a standardized variance of $\frac{1}{2}$, and if we assume that the allele frequency does not change over time, then neither does this quantity:

[Constant-freq] Allele frequencies do not change

The allele frequency p_k of each variant is the same across every generation. As a consequence, the standardization divides by a fixed quantity, and the standardized variance of each haploid allele is the same in every generation:

$$z_{ik} = \frac{a_{ik} - p_k}{\sqrt{2p_k(1 - p_k)}} \implies \text{Var}[z_{ik}] = \frac{\text{Var}[a_{ik}]}{2p_k(1 - p_k)} = \frac{p_k(1 - p_k)}{2p_k(1 - p_k)} = \frac{1}{2},$$

where $a_{ik} \in \{0, 1\}$ is the unstandardized haploid allele count and p_k is the allele frequency at variant k .

Commonly bundled into: no selection on the trait, no drift, no migration; in a finite population it is the large- N_e limit [7].

In contrast, the standardized diploid genotype can have non-unit variance and change over time if the two alleles are correlated, as will be demonstrated in the case of assortative mating. Our path diagram will therefore start with haploid alleles which are summed together to form diploid genotypes, which are then summed to form the additive genetic value, which is then added to the environmental value to form the phenotype.

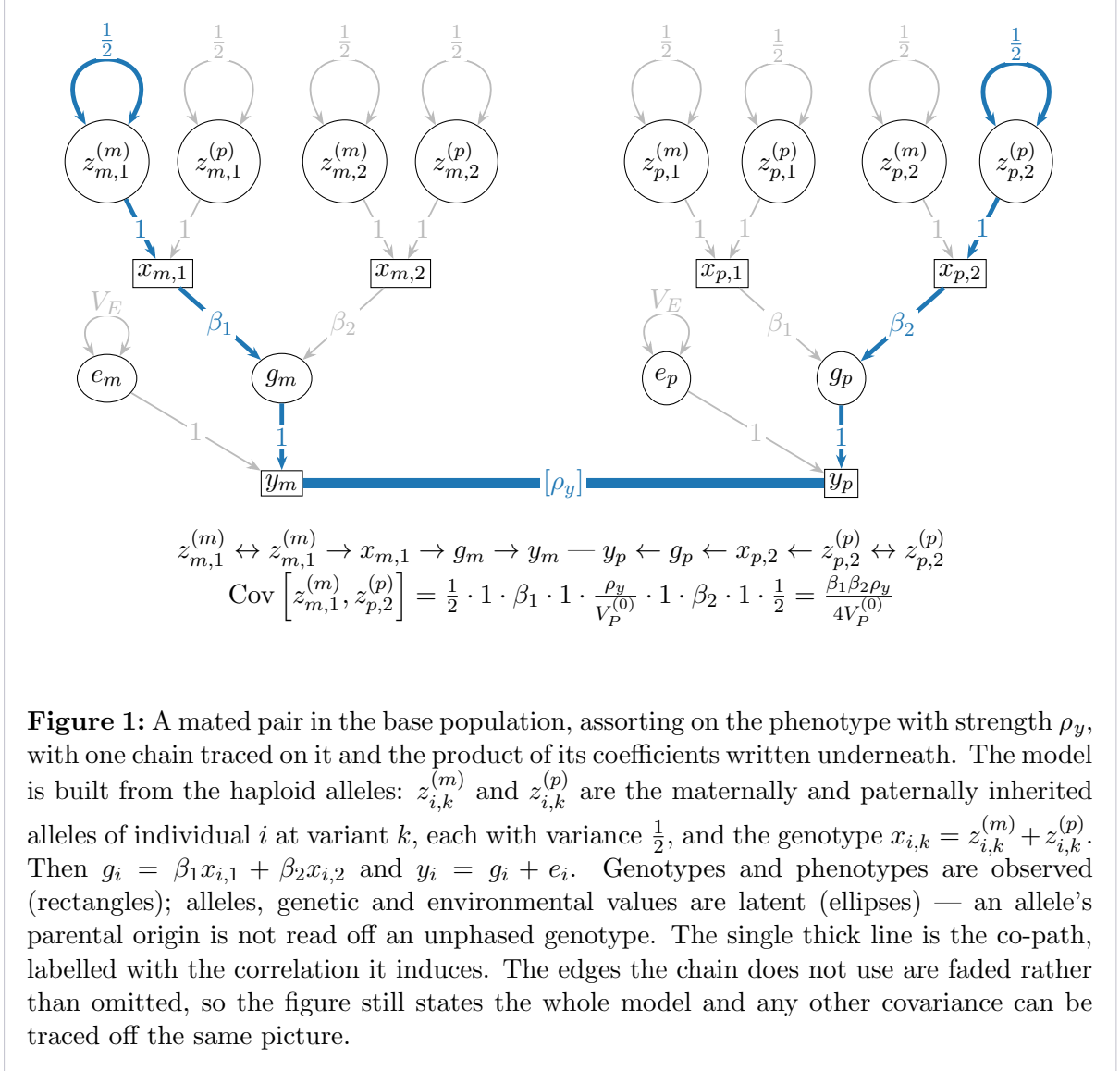
The diagram

Mates in generation $t = 0$ are formed according to [Primary-AM]. Below is the path diagram for a single pair that assortatively mated on the focal trait with strength ρ_y . The trait's genetic component is the additive sum of two causal variants, which is the smallest example that shows a *cross-variant* covariance; the formulas we state work for any number of variants.

One chain is traced on the diagram to show the rules in use. It runs from a single allele in one mate to a single allele at a *different* variant in the other — a covariance that is zero under random mating, and for which there is no common ancestor to appeal to. Multiplying the coefficients along it gives

$$\text{Cov}[z_{m,1}^{(m)}, z_{p,2}^{(p)}] = \frac{1}{2} \cdot \beta_1 \cdot \mu^{(0)} \cdot \beta_2 \cdot \frac{1}{2} = \frac{\beta_1 \beta_2 \rho_y}{4 V_P^{(0)}},$$

Because there are four such allele pairs behind any one pair of genotypes, they sum back to $\text{Cov}[x_{m,1}, x_{p,2}] = \beta_1 \beta_2 \rho_y / V_P^{(0)}$.



Doing the same for every pair of variables in the diagram gives the two tables below. Within an individual (Table 1) nothing has changed from the randomly mating case: these are base-population individuals, so their genotypes are still uncorrelated with each other and with their own environment.

	$x_{i,1}$	$x_{i,2}$	g_i	e_i	y_i
$x_{i,1}$	1	0	β_1	0	β_1
$x_{i,2}$	0	1	β_2	0	β_2
g_i	β_1	β_2	$V_A^{(0)}$	0	$V_A^{(0)}$
e_i	0	0	0	V_E	V_E
y_i	β_1	β_2	$V_A^{(0)}$	V_E	$V_P^{(0)}$

Table 1: Covariances within one individual $i \in \{m, p\}$, identical for both members of the pair. Here $V_A^{(0)} = \beta_1^2 + \beta_2^2$ and $V_P^{(0)} = V_A^{(0)} + V_E$. The last column — each variable's covariance with its own phenotype — is the only part of this table needed for computing the covariance between variables across mates.

The co-path induces covariance between the factors pertaining to mates such that:

$$\boxed{\text{Cov}[a_m, b_p] = \text{Cov}[a_m, y_m] \cdot \mu^{(0)} \cdot \text{Cov}[y_p, b_p]} \quad (12)$$

for any variables a and b belonging to the maternal and paternal member respectively. The entire cross-mate block is therefore the outer product of a single vector — the last column of Table 1 — scaled by $\mu^{(0)} = \rho_y/V_P^{(0)}$.

$\times \mu^{(0)}$	$x_{p,1}$	$x_{p,2}$	g_p	e_p	y_p
$x_{m,1}$	β_1^2	$\beta_1\beta_2$	$\beta_1V_A^{(0)}$	β_1V_E	$\beta_1V_P^{(0)}$
$x_{m,2}$	$\beta_1\beta_2$	β_2^2	$\beta_2V_A^{(0)}$	β_2V_E	$\beta_2V_P^{(0)}$
g_m	$\beta_1V_A^{(0)}$	$\beta_2V_A^{(0)}$	$(V_A^{(0)})^2$	$V_A^{(0)}V_E$	$V_A^{(0)}V_P^{(0)}$
e_m	β_1V_E	β_2V_E	$V_A^{(0)}V_E$	V_E^2	$V_EV_P^{(0)}$
y_m	$\beta_1V_P^{(0)}$	$\beta_2V_P^{(0)}$	$V_A^{(0)}V_P^{(0)}$	$V_EV_P^{(0)}$	$(V_P^{(0)})^2$

Table 2: Covariances between the two mates. **Every entry carries a factor $\mu^{(0)} = \rho_y/V_P^{(0)}$** , shown once in the corner rather than repeated in twenty-five cells. The block is symmetric, and is the outer product of $(\beta_1, \beta_2, V_A^{(0)}, V_E, V_P^{(0)})$ with itself — a direct consequence of (12). Setting $\rho_y = 0$ returns every entry to zero, recovering the randomly mating case of Section 1.

Three key results from Table 2 are highlighted here. We also generalize from two variants to M variants, where $g_i = \sum_{k=1}^M \beta_k x_{ik}$.

- $\text{Cov}[x_{mk}, x_{pl}] = \beta_k \beta_l \rho_y / V_P^{(0)} \neq 0$, for *every* pair of variants k and l . Alleles at different loci in different individuals are correlated, signed by the product of their effects and breaking the **[Non-IBD-indep]** assumption. Note that in the base population, the within-individual covariance $\text{Cov}[x_{mk}, x_{ml}]$ is still exactly zero for $k \neq l$; the disequilibrium exists only *between* the mates.
- $\text{Cov}[e_m, g_p] = V_A^{(0)} V_E \rho_y / V_P^{(0)} \neq 0$. One mate’s environment is correlated with the other’s genetic value, which is the break of **[Env-indep]** and **[No-IGE]** flagged at the start of this section. Despite one mate’s genotype not causally affecting the other’s environment, the two still become correlated due to the phenotypic matching.
- $\rho_g^{(0)} = \text{Corr}[g_m, g_p] = \frac{(V_A^{(0)})^2 \rho_y / V_P^{(0)}}{V_A^{(0)}} = \rho_y \frac{V_A^{(0)}}{V_P^{(0)}} = \rho_y h_{(0)}^2$. Mates’ genetic values are correlated according to the strength of assortative mating and the heritability of the sorting trait. Equivalently, this is the sum of the first result over all M^2 ordered pairs of variants, weighted by their effects: $\sum_k \sum_l \beta_k \beta_l \cdot \beta_k \beta_l \rho_y / V_P^{(0)} = (\sum_k \beta_k^2)^2 \rho_y / V_P^{(0)}$.

2.2.3 Consequences of assortative mating on first generation of offspring

Next, we will extend our path diagram to contain two offspring of an assortatively mated pair. For each variant, all offspring inherit one allele from each parent, but which allele they inherit from that parent (the maternal vs paternal allele) is uniformly random. For the purposes of tracking covariances, we model this by drawing an edge from each parental

allele to their respective inherited allele in the offspring and adding an exogenous source of variance. The directed edges have a coefficient of $\frac{1}{2}$ to reflect that the expected inherited allele in the offspring is the average of the two alleles of the parent being inherited from.

Write o for an offspring of the mated pair, and let $B_{ok}^{(m)} \in \{0, 1\}$ indicate which of the mother's two alleles was transmitted at variant k . We arbitrarily label the maternal allele $z_{mk}^{(m)}$ and the paternal allele $z_{mk}^{(p)}$ such that if $B_{ok}^{(m)} = 1$, then the offspring inherits the maternal allele, and if $B_{ok}^{(m)} = 0$ then the offspring inherits the paternal allele.

$$z_{ok}^{(m)} = B_{ok}^{(m)} z_{mk}^{(m)} + (1 - B_{ok}^{(m)}) z_{mk}^{(p)}, \quad B_{ok}^{(m)} \sim \text{Bernoulli}(\tfrac{1}{2}), \quad B_{ok}^{(m)} \perp z_{mk}. \quad (13)$$

A path diagram represents the *linear projection* of the random variable: the best prediction of the offspring's allele from the mother's two, which is their average,

$$\mathbb{E}[z_{ok}^{(m)} \mid z_{mk}^{(m)}, z_{mk}^{(p)}] = \tfrac{1}{2} z_{mk}^{(m)} + \tfrac{1}{2} z_{mk}^{(p)} = \tfrac{1}{2} x_{mk}. \quad (14)$$

The exogenous variance is what the projection leaves behind. Subtracting (14) from (13) gives the *segregation deviation*

$$\epsilon_{ok}^{(m)} := z_{ok}^{(m)} - \tfrac{1}{2} x_{mk} = (B_{ok}^{(m)} - \tfrac{1}{2})(z_{mk}^{(m)} - z_{mk}^{(p)}), \quad (15)$$

which is mean-zero and uncorrelated with both of the mother's alleles — a disturbance in the sense of Section 2.2.1, and drawn as one. The offspring's paternally inherited allele $z_{ok}^{(p)}$ is built the same way from the father's two alleles, with its own independent $B_{ok}^{(p)}$ and residual $\epsilon_{ok}^{(p)}$.

Covariance between alleles

We now define the covariances that will be repeatedly used in deriving the consequences of assortative mating. For an individual i of generation t and variants k and l where $u, v \in \{m, p\}$,

$$\boxed{a_{kl}^{(t)} := \text{Cov}[z_{ik}^{(u)}, z_{il}^{(v)}] \quad (u \neq v), \quad s_{kl}^{(t)} := \text{Cov}[z_{ik}^{(u)}, z_{il}^{(u)}]} \quad (16)$$

In words, a denotes the covariance between alleles *across* gametes, while s denotes the covariance between alleles on the *same* gamete. When $k = l$, $s_{kk}^{(t)}$ is the variance of an individual allele, which is fixed for all generations due to how we standardized variables according to [\[Constant-freq\]](#):

$$s_{kk}^{(t)} \equiv \tfrac{1}{2} \quad \text{for all } t, \quad (17)$$

We can also write the covariance between two genotypes in the same individual as the sum of four allelic covariances:

$$\text{Cov}[x_{ik}, x_{il}] = \sum_{u, v \in \{m, p\}} \text{Cov}[z_{ik}^{(u)}, z_{il}^{(v)}] = 2(s_{kl}^{(t)} + a_{kl}^{(t)}), \quad (18)$$

In the case of $k = l$, it becomes the variance of genotypes, since $s_{kk}^{(t)} \equiv \frac{1}{2}$:

$$\begin{aligned} \text{Var}[x_{ik}] &= 2(s_{kk}^{(t)} + a_{kk}^{(t)}) \\ \boxed{\text{Var}[x_{ik}] &= 1 + 2a_{kk}^{(t)}} \end{aligned} \quad (19)$$

The third quantity we need is the covariance between a single allele and the phenotype of the individual carrying it. Expanding the phenotype into its causes and then each genotype into its two alleles,

$$c_k^{(t)} := \text{Cov}[z_{ik}^{(u)}, y_i] = \sum_{l=1}^M \beta_l \text{Cov}[z_{ik}^{(u)}, x_{il}] = \sum_{l=1}^M \beta_l (s_{kl}^{(t)} + a_{kl}^{(t)}) \quad (20)$$

The first step drops e_i , which is uncorrelated with every allele of the same individual by **[GE-indep]**. The second splits each genotype x_{il} into the allele sitting on the same gamete as $z_{ik}^{(u)}$, contributing $s_{kl}^{(t)}$, and the one on the opposite gamete, contributing $a_{kl}^{(t)}$. A single allele therefore reaches the phenotype through *every* variant the individual carries, not only through its own.

Summing over alleles in the same way gives the additive variance of the generation:

$$\begin{aligned} V_A^{(t)} &= \text{Var}\left[\sum_k \beta_k x_{ik}\right] = \sum_k \sum_l \beta_k \beta_l \text{Cov}[x_{ik}, x_{il}] \\ &= 2 \sum_k \sum_l \beta_k \beta_l (s_{kl}^{(t)} + a_{kl}^{(t)}) \end{aligned} \quad (21)$$

$$V_A^{(t)} = 2 \sum_k \beta_k c_k^{(t)}$$

with $V_P^{(t)} = V_A^{(t)} + V_E$, since assortment on the phenotype does not change the environmental variance. Similarly, we define generation-specific heritability as follows:

$$h_{(t)}^2 := \frac{V_A^{(t)}}{V_P^{(t)}} = \frac{V_A^{(t)}}{V_A^{(t)} + V_E} = \frac{2 \sum_k \beta_k c_k^{(t)}}{2 \sum_k \beta_k c_k^{(t)} + V_E} \quad (22)$$

We'll also confirm here that $\rho_g^{(t)} = \frac{\rho_y V_A^{(t)}}{V_P^{(t)}}$. A single co-path joins the two mates, so (12) applies with $\mu^{(t)}$ in place of $\mu^{(0)}$, and a genetic value's covariance with its own carrier's phenotype is $\text{Cov}[g_i, g_i + e_i] = V_A^{(t)}$ by **[GE-indep]**. Both mates have genetic variance $V_A^{(t)}$, so the denominator of the correlation is $V_A^{(t)}$ as well:

$$\rho_g^{(t)} = \frac{\text{Cov}[g_m, g_p]}{\sqrt{\text{Var}[g_m] \text{Var}[g_p]}} = \frac{\overbrace{\text{Cov}[g_m, y_m]}^{V_A^{(t)}} \cdot \mu^{(t)} \cdot \overbrace{\text{Cov}[y_p, g_p]}^{V_A^{(t)}}}{V_A^{(t)}} = \mu^{(t)} V_A^{(t)} \quad (23)$$

$$\rho_g^{(t)} = \frac{\rho_y V_A^{(t)}}{V_P^{(t)}} = \rho_y h_{(t)}^2$$

The genetic correlation between mates is the phenotypic one scaled by that generation's heritability, exactly as it was in the base population.

We can also now write the variance of the segregation deviation. The indicator $B_{ok}^{(m)}$ takes values in $\{0, 1\}$, so $(B_{ok}^{(m)} - \frac{1}{2})^2 = \frac{1}{4}$ with probability one, and it is independent

of the alleles it selects between. Squaring (15) and taking expectations therefore factors cleanly:

$$\text{Var}[\epsilon_{ok}^{(m)}] = \mathbb{E}\left[\left(B_{ok}^{(m)} - \frac{1}{2}\right)^2\right] \text{Var}[z_{mk}^{(m)} - z_{mk}^{(p)}] = \frac{1}{4} \left(\underbrace{\frac{1}{2}}_{s_{kk}^{(t)}} + \underbrace{\frac{1}{2}}_{s_{kk}^{(t)}} - 2 \underbrace{a_{kk}^{(t)}}_{\text{Cov}[z_{mk}^{(m)}, z_{mk}^{(p)}]} \right), \quad (24)$$

where the parent is of generation t . Collecting terms,

$$\boxed{\text{Var}[\epsilon_{ok}^{(m)}] = \frac{1}{4} - \frac{1}{2} a_{kk}^{(t)}} \quad (25)$$

So a parent whose two alleles are more alike contributes *less* segregation variance: it matters less which of the two was transmitted.

First generation of offspring

In a generation-0 individual, their own alleles are uncorrelated with each other. At the same variant, the two alleles are non-IBD by [\[No-inbreeding\]](#) and hence independent by [\[Non-IBD-indep\]](#); at different variants, the base population is in linkage equilibrium, which is the within-individual zero already noted below Table 2:

$$\boxed{a_{kl}^{(0)} = 0 \text{ for all } k, l, \quad s_{kl}^{(0)} = 0 \text{ } (k \neq l), \quad c_k^{(0)} = \frac{1}{2}\beta_k} \quad (26)$$

The third follows from the first two: with every off-diagonal term of (20) equal to zero, only $l = k$ survives, leaving $\beta_k s_{kk}^{(0)} = \frac{1}{2}\beta_k$. Setting $a_{kk}^{(0)} = 0$ in (19) recovers $\text{Var}[x_{ik}] = 1$, and in (25) recovers the familiar segregation variance of $\frac{1}{4}$.

In a generation-1 individual, this is no longer the case, which can be seen by the below path diagram of two generation-1 offspring of a mated pair that assortatively mated on the focal trait with strength ρ_y , for a trait with two causal variants.

We can use our path tracing rules to compute $a_{kl}^{(1)}$. Exactly four chains connect an allele of the offspring to an allele on its *other* gamete, at variants k and l . Each traces back from $z_{ok}^{(m)}$ to *one* of the mother's two alleles at variant k , turns around on that allele's variance, runs up through her genotype and phenotype, crosses the co-path, and comes back down the father's side to *one* of his two alleles at variant l and into $z_{ol}^{(p)}$. Two choices on each side give $2 \times 2 = 4$ chains identical in value, one for each $u, v \in \{m, p\}$:

$$\begin{aligned} & z_{ok}^{(m)} \leftarrow z_{mk}^{(u)} \leftrightarrow z_{mk}^{(u)} \rightarrow x_{mk} \rightarrow g_m \rightarrow y_m \\ \text{--- } & y_p \leftarrow g_p \leftarrow x_{pl} \leftarrow z_{pl}^{(v)} \leftrightarrow z_{pl}^{(v)} \rightarrow z_{ol}^{(p)}, \end{aligned}$$

Multiplying each coefficient along one chain returns:

$$\underbrace{\frac{1}{2}}_{\text{transmission}} \cdot \underbrace{\frac{1}{2}}_{\text{Var}[z]} \cdot \beta_k \cdot \mu^{(0)} \cdot \beta_l \cdot \underbrace{\frac{1}{2}}_{\text{Var}[z]} \cdot \underbrace{\frac{1}{2}}_{\text{transmission}} = \frac{\beta_k \beta_l \mu^{(0)}}{16}.$$

Summing across the four chains and expanding $\mu^{(0)}$,

$$\boxed{a_{kl}^{(1)} = \frac{\beta_k \beta_l \rho_y}{4 V_P^{(0)}}} \quad (27)$$

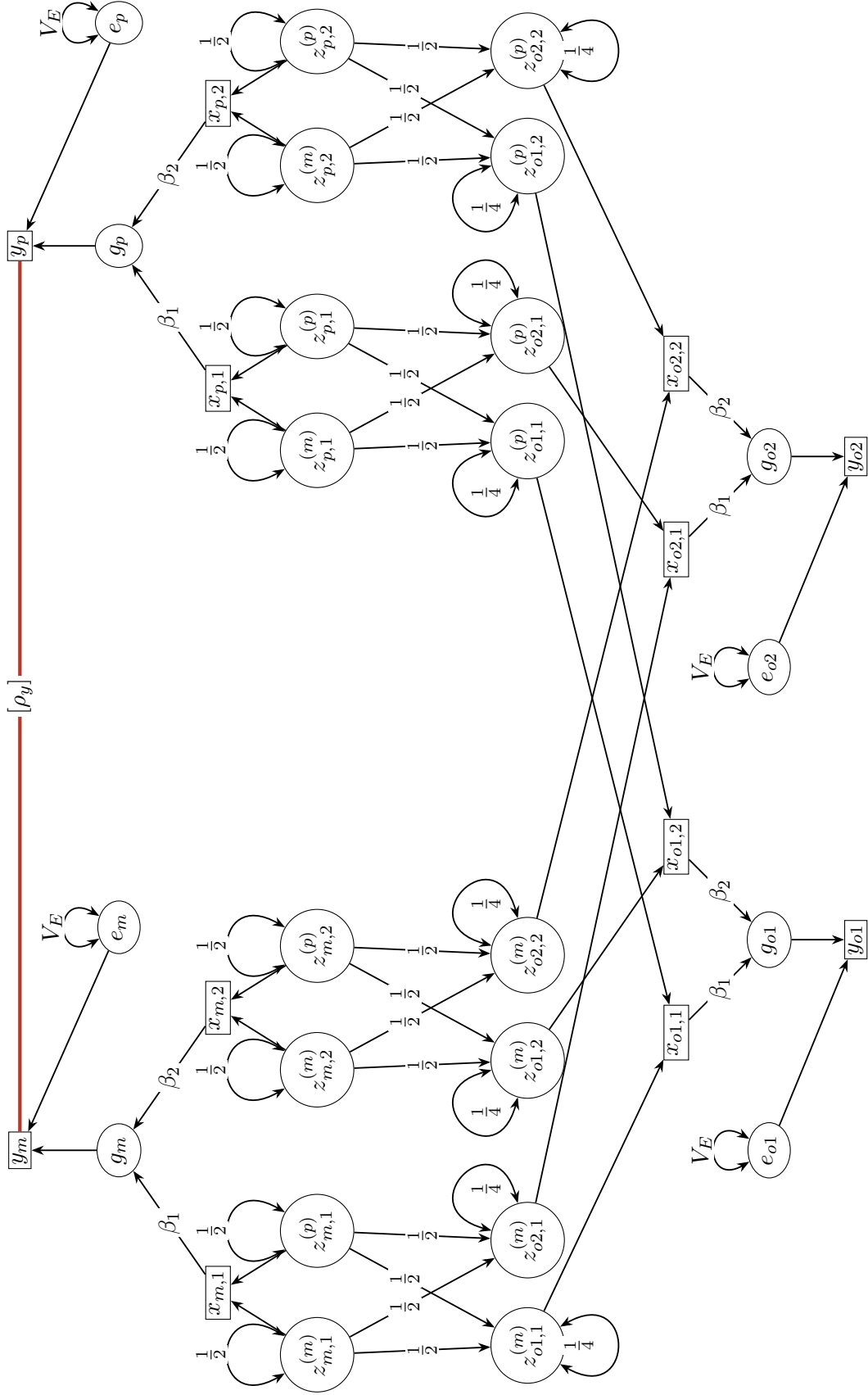


Figure 2: One mated pair from the base population and two of their offspring, at the allele level, for a trait with two causal variants. Each inherited allele is fed by *both* alleles of the parent it came from, at $\frac{1}{2}$ each, per (14). The self-loop of $\frac{1}{4}$ on each offspring allele is its segregation variance, (25) evaluated at $a_{kk}^{(0)} = 0$.

The same-gamete companion is easier: two alleles the offspring received from the *same* parent at different variants have no chain at all connecting them, because that parent's own alleles are uncorrelated across variants due to them being of the base population, so

$$\boxed{s_{kl}^{(1)} = 0 \quad (k \neq l)} \quad (28)$$

One generation of assortment therefore produces association *across* an individual's two gametes and none within either of them.

Rearranging (23) gives $\rho_y/V_P^{(0)} = \rho_g^{(0)}/V_A^{(0)}$, which lets (27) be restated in terms of how strongly the parents' genetic values were correlated rather than their phenotypes:

$$\boxed{a_{kl}^{(1)} = \frac{\beta_k \beta_l}{4 V_A^{(0)}} \rho_g^{(0)}} \quad (29)$$

The association between two of the offspring's alleles is the product of their effect sizes, taken as a share of the additive variance, scaled by the genetic correlation between its parents.

Feeding (27) and (28) into (20) gives the first-generation allele-phenotype covariance. Only $l = k$ survives from $s^{(1)}$, while every l contributes through $a^{(1)}$:

$$c_k^{(1)} = \underbrace{\frac{1}{2}\beta_k}_{l=k \text{ term of } s^{(1)}} + \sum_l \beta_l \frac{\beta_k \beta_l \rho_y}{4 V_P^{(0)}} = \frac{1}{2}\beta_k \left(1 + \frac{\rho_y V_A^{(0)}}{2 V_P^{(0)}} \right) \quad (30)$$

$$\boxed{c_k^{(1)} = \frac{1}{2}\beta_k \left(1 + \frac{1}{2}\rho_g^{(0)} \right)}$$

using $\sum_l \beta_l^2 = V_A^{(0)}$. Every variant's initial covariance with the phenotype is inflated by the same factor.

We use the path tracing rules to compute the covariance between an individual and a full sibling and a parent, shown in Table 3. Every entry there turns out to be its random-mating value plus a term proportional to $\rho_g^{(0)}$.

	self	full siblings	parent–offspring
genotypes, $k = l$	$1 + \frac{\beta_k^2}{2V_A^{(0)}}\rho_g^{(0)}$	$\frac{1}{2} + \frac{\beta_k^2}{2V_A^{(0)}}\rho_g^{(0)}$	$\frac{1}{2} + \frac{\beta_k^2}{2V_A^{(0)}}\rho_g^{(0)}$
genotypes, $k \neq l$	$\frac{\beta_k\beta_l}{2V_A^{(0)}}\rho_g^{(0)}$	$\frac{\beta_k\beta_l}{2V_A^{(0)}}\rho_g^{(0)}$	$\frac{\beta_k\beta_l}{2V_A^{(0)}}\rho_g^{(0)}$
genetic values	$V_A^{(0)}(1 + \frac{1}{2}\rho_g^{(0)})$	$\frac{1}{2}V_A^{(0)}(1 + \rho_g^{(0)})$	$\frac{1}{2}V_A^{(0)}(1 + \rho_g^{(0)})$
phenotypes	$V_A^{(0)}(1 + \frac{1}{2}\rho_g^{(0)}) + V_E$	$\frac{1}{2}V_A^{(0)}(1 + \rho_g^{(0)})$	$\frac{1}{2}V_A^{(0)}(1 + \rho_y)$

Table 3: Variances and covariances for an individual of the **first generation produced under assortative mating**, against itself and against each kind of first-degree relative: the “self” column is the pair (o_1, o_1) , “full siblings” is (o_1, o_2) , and “parent–offspring” is (m, o_1) . The first row is a genotype against a genotype at the *same* variant ($k = l$), the second at two *different* variants. Every entry is its random-mating value plus **what one generation of assortment added**. Writing the added terms through $\rho_g^{(0)}$ rather than $\rho_y/V_P^{(0)}$ leaves the two relationships differing in a single symbol: siblings carry $\rho_g^{(0)}$ at the phenotypic level where parent and offspring carry ρ_y .

Note that covariances for parents vs full-siblings only differ at the phenotypic level due to one extra non-zero term that emerges for direct descendants:

$$\begin{aligned}
\text{Cov}[y_m, y_{o_1}] &= \underbrace{\text{Cov}[g_m, g_{o_1}]}_{\frac{1}{2}V_A^{(0)} + \frac{\rho_y(V_A^{(0)})^2}{2V_P^{(0)}}} + \underbrace{\text{Cov}[e_m, g_{o_1}]}_{\frac{\rho_y V_A^{(0)} V_E}{2V_P^{(0)}}} \\
&= \frac{1}{2}V_A^{(0)} + \frac{\rho_y V_A^{(0)}(V_A^{(0)} + V_E)}{2V_P^{(0)}} = \frac{1}{2}V_A^{(0)}(1 + \rho_y),
\end{aligned}$$

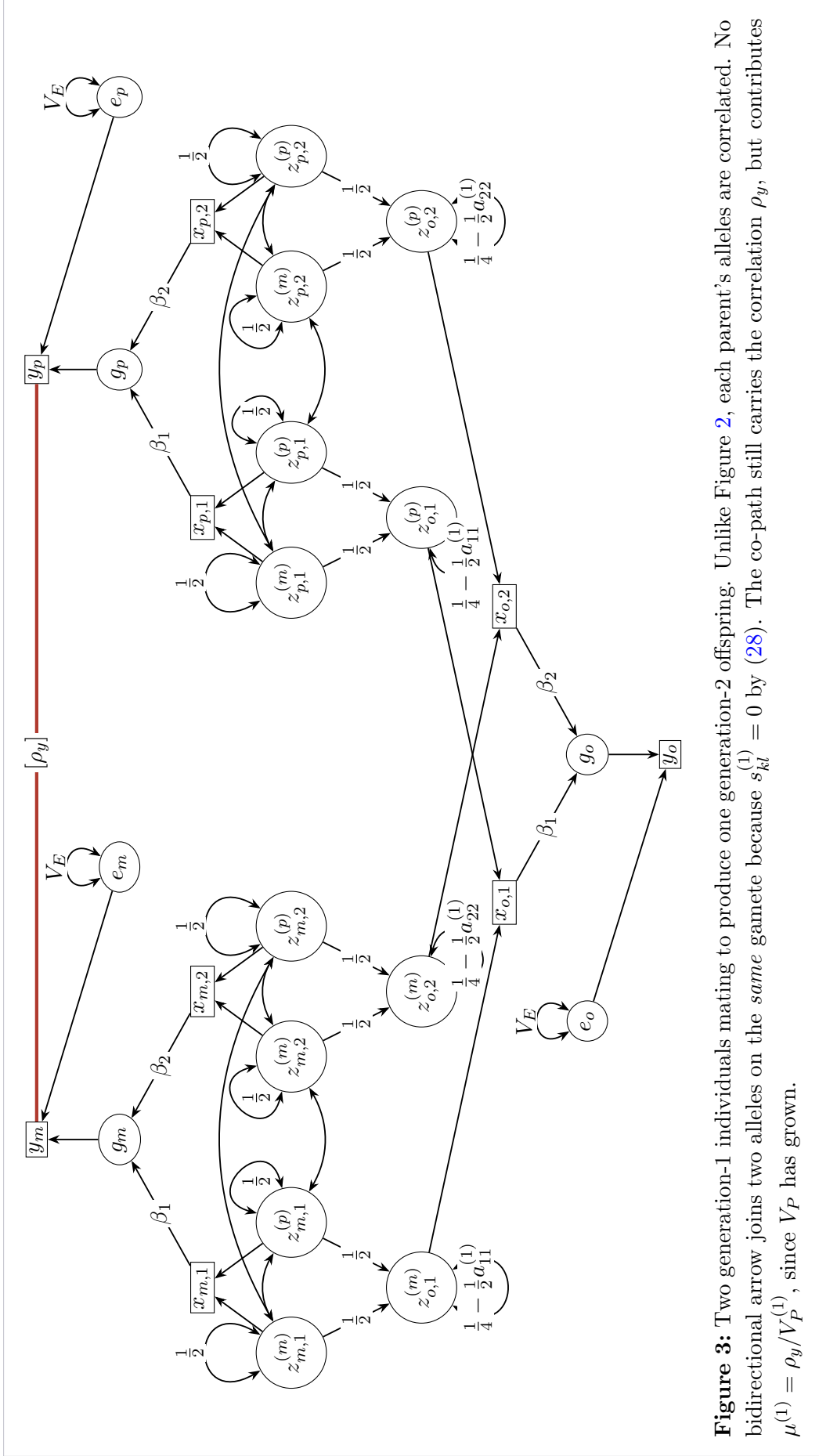
where the last step uses $V_P^{(0)} = V_A^{(0)} + V_E$. A parent’s environment is correlated with their child’s genetic value — the child carries half the alleles of the assortatively chosen partner, whose genotype is correlated with this parent’s environment.

2.2.4 Compounding across generations

If we now produce a second generation of offspring, the effects of another generation of assortative mating compound on the existing effects of past assortative mating. Namely, unlike before, the two alleles that a parent carries are now correlated according to their effect size.

It would be unwieldy to make a diagram for three generations. Instead we summarize the base population by its impact on the covariance of generation-1 alleles.

Covariance across gametes. We still think in terms of the four chains described in (27), but now any chain connecting a parent’s allele up to its own phenotype is no longer just $\frac{1}{2}\beta_k$: the parent’s alleles are correlated now, so a chain leaving $z_{ik}^{(u)}$ can reach



the phenotype through *any* other allele the parent carries. Every such route is already collected in $c_k^{(t)}$, which (20) defined as exactly that covariance. The four chains then contribute $c_k^{(t)} \cdot \mu^{(t)} \cdot c_l^{(t)}$ in total, so

$$\boxed{a_{kl}^{(t+1)} = \mu^{(t)} c_k^{(t)} c_l^{(t)}} \quad (31)$$

Setting $t = 0$ recovers (27), since $c_k^{(0)} = \frac{1}{2}\beta_k$ and $\mu^{(0)} = \rho_y/V_P^{(0)}$.

Covariance within a gamete. When both alleles came from the *same* parent, no chain can leave that parent and therefore the co-path is not crossed:

$$\boxed{s_{kl}^{(t+1)} = \frac{1}{4}\text{Cov}[x_{ik}, x_{il}] = \frac{1}{2}\left(s_{kl}^{(t)} + a_{kl}^{(t)}\right)} \quad (32)$$

using (18) for the parent's genotype covariance. In other words, when picking two non-linked alleles from the same gamete in an offspring, they are equally likely to have been from the same vs different gamete in the parent. Therefore, the covariance between the two offspring's alleles is the average of the across- vs same-gamete covariance in the parent generation.

While $s_{kl}^{(1)} = 0$, the second generation is where the same-gamete covariance becomes non-zero: $s_{kl}^{(2)} = \frac{1}{2}a_{kl}^{(1)} = \beta_k\beta_l\rho_y/(8V_P^{(0)})$.

Covariance between an allele and the phenotype. We rewrite the formula for c from (20), one generation later, as two separate sums:

$$c_k^{(t+1)} = \sum_{l=1}^M \beta_l \left(s_{kl}^{(t+1)} + a_{kl}^{(t+1)} \right) = \underbrace{\sum_l \beta_l a_{kl}^{(t+1)}}_{\text{across gametes}} + \underbrace{\sum_l \beta_l s_{kl}^{(t+1)}}_{\text{same gamete}}. \quad (33)$$

The across-gamete part can be written in terms of the previous generation's c by (31). After pulling out common terms, we are left with a summation that can be simplified according to (21). We then plug in $\mu^{(t)} = \rho_y/V_P^{(t)}$, which leaves $\rho_y V_A^{(t)}/V_P^{(t)}$ — and that is $\rho_g^{(t)}$ by (23):

$$\begin{aligned} \sum_l \beta_l a_{kl}^{(t+1)} &= \sum_l \beta_l \mu^{(t)} c_k^{(t)} c_l^{(t)} = \mu^{(t)} c_k^{(t)} \underbrace{\sum_l \beta_l c_l^{(t)}}_{V_A^{(t)}/2} = \frac{\rho_y V_A^{(t)}}{2 V_P^{(t)}} c_k^{(t)} \\ &= \frac{1}{2} \rho_g^{(t)} c_k^{(t)}, \end{aligned} \quad (34)$$

For the same-gamete part we fill in the recursive expression for $s_{kl}^{(t+1)}$ according to (32), which lets us write it in terms of $c_k^{(t)}$ according to (20). However, we must be careful to

replace the $k = l$ case with its fixed value of $s_{kk}^{(t)} \equiv \frac{1}{2}$:

$$\begin{aligned} \sum_l \beta_l s_{kl}^{(t+1)} &= \underbrace{\frac{1}{2} \sum_l \beta_l (s_{kl}^{(t)} + a_{kl}^{(t)})}_{c_k^{(t)}} + \beta_k \left[\underbrace{\frac{1}{2}}_{s_{kk}^{(t+1)}} - \underbrace{\frac{1}{2}(\frac{1}{2} + a_{kk}^{(t)})}_{\text{what (32) would give for } k=l} \right] \\ &= \frac{1}{2} c_k^{(t)} + \beta_k \left(\frac{1}{4} - \frac{1}{2} a_{kk}^{(t)} \right). \end{aligned} \quad (35)$$

The bracket is the correction for the one term the recursion does not govern.

We now add the two halves. Note that $\rho_g^{(t)}$ appears in only one of them: it enters through (34), because that is the half whose chains cross the co-path between the parents, and it is absent from (35), whose chains stay inside one parent. What the two halves share is a factor of $\frac{1}{2} c_k^{(t)}$, and collecting that factor is what puts $\rho_g^{(t)}$ next to a 1:

$$\begin{aligned} c_k^{(t+1)} &= \underbrace{\frac{1}{2} \rho_g^{(t)} c_k^{(t)}}_{\text{across gametes, (34)}} + \underbrace{\frac{1}{2} c_k^{(t)} + \beta_k \left(\frac{1}{4} - \frac{1}{2} a_{kk}^{(t)} \right)}_{\text{same gamete, (35)}} \\ &\boxed{c_k^{(t+1)} = \underbrace{\frac{1}{2} (1 + \rho_g^{(t)}) c_k^{(t)}}_{\text{inherited}} + \underbrace{\beta_k \left(\frac{1}{4} - \frac{1}{2} a_{kk}^{(t)} \right)}_{\text{Var}[\epsilon_{ok}^{(m)}]}} \end{aligned} \quad (36)$$

So the 1 is the allele the offspring actually received, and the $\rho_g^{(t)}$ beside it is the extra covariance that allele picked up because the parent it came from was chosen to resemble the other parent. Under random mating $\rho_g^{(t)} = 0$ and only the 1 remains. We can notice that the segregation variance (25) naturally emerges from our formula and brings with it a satisfying interpretation. We have previously thought of offspring alleles as having a directly-inherited component from the parent (equivalent to half the parent's genotype) plus segregation deviation that is uncorrelated with the parent's genotype. The former contributes half the covariance present in the parent generation, but amplified slightly due to assortative mating predicting the total covariance of correlated variants coming from the other parent. The latter contributes additional variance directly proportional to that variant's effect size.

We can expand out the terms of equation (36) such that they only rely on the starting parameters — the effect sizes β_k , the phenotypic correlation between mates ρ_y , and the environmental variance V_E — and previous states of c . Substituting $\rho_g^{(t)} = \rho_y V_A^{(t)} / V_P^{(t)}$, then $V_A^{(t)} = 2 \sum_j \beta_j c_j^{(t)}$ and $V_P^{(t)} = V_A^{(t)} + V_E$, and finally $a_{kk}^{(t)} = \mu^{(t-1)} (c_k^{(t-1)})^2$ leaves

$$c_k^{(t+1)} = \frac{1}{4} \beta_k + \frac{1}{2} \left(1 + \frac{2 \rho_y \sum_j \beta_j c_j^{(t)}}{2 \sum_j \beta_j c_j^{(t)} + V_E} \right) c_k^{(t)} - \frac{\rho_y \beta_k}{2 \left(2 \sum_j \beta_j c_j^{(t-1)} + V_E \right)} (c_k^{(t-1)})^2, \quad (37)$$

with $c_k^{(t)} = \frac{1}{2} \beta_k$ for every $t \leq 0$, the base population having been reached without assortment.

This reveals that the covariances in this population follow an *autonomous second-order difference equation*: autonomous because the update rule is the same at every generation,

and second-order because it reaches back two of them. The M variants form one coupled system rather than M separate equations, since each variant's update reads all the others through $V_A^{(t)}$. It is also *nonlinear* — $c_k^{(t-1)}$ appears squared, and $c^{(t)}$ appears inside a denominator — so the usual closed-form solutions for linear recursions do not apply, and we look instead for a *fixed point*: a c that the update leaves unchanged. The leading factor $\frac{1}{2}(1 + \rho_g^{(t)})$ is less than one whenever $\rho_g^{(t)} < 1$, which makes the update a contraction near such a point, so the approach to it is geometric rather than oscillatory or divergent: the remaining distance falls by a roughly constant fraction each generation.

		$t \leq 0$ base pop.	$t = 1$	general (t)
allele-phenotype	$c_k^{(t)}$	$\frac{1}{2}\beta_k$	$\frac{1}{2}\beta_k(1 + \frac{1}{2}\rho_g^{(0)})$	$\frac{1}{2}(1 + \rho_g^{(t-1)})c_k^{(t-1)} + \beta_k(\frac{1}{4} - \frac{1}{2}a_{kk}^{(t-1)})$
across gametes	$a_{kl}^{(t)}$	0	$\frac{\beta_k\beta_l}{4V_A^{(0)}}\rho_g^{(0)}$	$\mu^{(t-1)}c_k^{(t-1)}c_l^{(t-1)}$
same gamete	$s_{kl}^{(t)}, k \neq l$	0	0	$\frac{1}{2}(s_{kl}^{(t-1)} + a_{kl}^{(t-1)})$
genotype variance	$\text{Var}[x_{ik}]$	1	$1 + \frac{\beta_k^2}{2V_A^{(0)}}\rho_g^{(0)}$	$1 + 2a_{kk}^{(t)}$
segregation variance	$\text{Var}[\epsilon_{ok}^{(m)}]$	$\frac{1}{4}$	$\frac{1}{4} - \frac{\beta_k^2}{8V_A^{(0)}}\rho_g^{(0)}$	$\frac{1}{4} - \frac{1}{2}a_{kk}^{(t)}$
additive variance	$V_A^{(t)}$	$V_A^{(0)}$	$V_A^{(0)}(1 + \frac{1}{2}\rho_g^{(0)})$	$2\sum_k \beta_k c_k^{(t)}$
phenotypic variance	$V_P^{(t)}$	$V_A^{(0)} + V_E$	$V_A^{(1)} + V_E$	$V_A^{(t)} + V_E$
heritability	$h_{(t)}^2$	$V_A^{(0)}/V_P^{(0)}$	$V_A^{(1)}/V_P^{(1)}$	$V_A^{(t)}/V_P^{(t)}$
genetic correlation	$\rho_g^{(t)}$	$\rho_y h_{(0)}^2$	$\rho_y h_{(1)}^2$	$\rho_y h_{(t)}^2$
co-path coefficient	$\mu^{(t)}$	$\rho_y/V_P^{(0)}$	$\rho_y/V_P^{(1)}$	$\rho_y/V_P^{(t)}$

Table 4: Every quantity derived in this section, from the base population forward, for a trait with M variants. Every column gives that generation's own value, the left-hand side always being the quantity named in the second column. The three recursions therefore reach back to $t - 1$ in the last column, while the rest are evaluated at t directly. The first block are the three allele covariances, of which only c is a recursion the others do not feed: a and s are read off it, and $s_{kk}^{(t)} \equiv \frac{1}{2}$. The second block is read off the same generation's c , and involves no recursion at all. The base-population column is written $t \leq 0$ because nothing has happened yet: its ancestors mated at random, so c , a and s take these values at every $t \leq 0$, which is what lets the second-order (37) start. Assortment begins with the generation-0 pairing, so $\rho_g^{(t)}$ and $\mu^{(t)}$ in that column are the ones governing the step to $t = 1$.

2.2.5 Equilibrium

Our next goal is to solve for the values that lead to an equilibrium state in the recursive covariance formulas we expressed above. An equilibrium is a state the update in (36) leaves unchanged, so we look for a c satisfying $c_k^{(t+1)} = c_k^{(t)} = c_k^{(t-1)}$, which we write $c_k^{(\text{eq})}$.

[AM-equilibrium] Intergenerational equilibrium

Assortment has been operating at the same strength for long enough that the covariances have stopped changing from one generation to the next.

Commonly bundled into: equilibrium assortative mating; reached after roughly six to ten generations [8].

Under equilibrium, we fill in the same value, $c_k^{(eq)}$ for all states of c_k in (36) after replacing $a_{kk}^{(eq)} = \mu^{(eq)} (c_k^{(eq)})^2$.

$$\begin{aligned}
c_k^{(eq)} &= \frac{1}{2} (1 + \rho_g^{(eq)}) c_k^{(eq)} + \beta_k \left(\frac{1}{4} - \frac{1}{2} \mu^{(eq)} (c_k^{(eq)})^2 \right) \\
\frac{1}{2} (1 - \rho_g^{(eq)}) c_k^{(eq)} &= \frac{1}{4} \beta_k - \frac{1}{2} \mu^{(eq)} \beta_k (c_k^{(eq)})^2 \\
2(1 - \rho_g^{(eq)}) c_k^{(eq)} &= \beta_k - 2 \mu^{(eq)} \beta_k (c_k^{(eq)})^2 \\
2\mu^{(eq)} \beta_k (c_k^{(eq)})^2 + 2(1 - \rho_g^{(eq)}) c_k^{(eq)} - \beta_k &= 0
\end{aligned} \tag{38}$$

$$c_k^{(eq)} = \frac{-(1 - \rho_g^{(eq)}) \pm \sqrt{(1 - \rho_g^{(eq)})^2 + 2 \mu^{(eq)} \beta_k^2}}{2 \mu^{(eq)} \beta_k} \tag{39}$$

$$\boxed{c_k^{(eq)} = \frac{\beta_k}{(1 - \rho_g^{(eq)}) + \sqrt{(1 - \rho_g^{(eq)})^2 + 2 \mu^{(eq)} \beta_k^2}}}$$

Only the $+$ root is admissible: the $-$ root gives a $c_k^{(eq)}$ of the opposite sign to β_k , and does not reduce to the base-population value when assortment is switched off.

However, we still have the problem that $\mu^{(eq)}$ and $\rho_g^{(eq)}$ depend on $V_A^{(eq)}$, which itself depends on the sum of $c_k^{(eq)}$ for all k , which are quantities we are solving for. Summing (39) over variants does not resolve this, because each variant carries its own square root containing its own $2\mu^{(eq)}\beta_k^2$, so the M terms cannot be cleanly gathered into one. To make obtaining a closed form solution manageable, we need to apply an approximation that is justified by the following assumptions:

[Large-Me] The additive variance is spread across many variants

Taking inspiration from O'Connor et al. [9], define the *effective number of variants* as

$$M_e := \frac{(V_A^{(0)})^2}{\sum_k \beta_k^4} = \frac{(\sum_k \beta_k^2)^2}{\sum_k \beta_k^4} \tag{40}$$

This is closely related to the same quantity defined in O'Connor et al. [9], albeit ours is smaller by a factor of three because we are aiming for the property that $M_e = M$ when all variants have the same effect size, rather than when all variants have the same *Gaussian prior*.

Specifically, we assume that the effective number of variants is large:

$$M_e \gg 1.$$

Equivalently, no single variant explains an appreciable share of the additive variance. Since $\sum_j \beta_j^4 \geq \beta_k^4$ for any single k , rearranging (40) gives

$$\frac{\beta_k^2}{V_A^{(0)}} \leq \frac{1}{\sqrt{M_e}} \quad \text{for every } k, \quad (41)$$

so $M_e \gg 1$ forces every variance share below $M_e^{-1/2} \ll 1$.

Commonly bundled into: the polygenic, or infinitesimal, model with large M ; M_e is estimable from summary statistics [9].

[Moderate-AM] Assortment is not maximal

The genetic correlation between mates stays bounded away from its ceiling, $\rho_g^{(\text{eq})} < 1$, so that $1 - \rho_g^{(\text{eq})}$ is not close to zero.

Commonly bundled into: any trait that is not both highly heritable and highly assorted.

[Uniform-inflation] Assortment inflates every variant's covariance with the phenotype alike

$$c_k^{(\text{eq})} = \frac{\beta_k}{(1 - \rho_g^{(\text{eq})}) + \sqrt{(1 - \rho_g^{(\text{eq})})^2 + \underbrace{2\mu^{(\text{eq})}\beta_k^2}_{\text{negligible}}} \quad (42)$$

$$\boxed{c_k^{(\text{eq})} \approx \frac{\beta_k}{2(1 - \rho_g^{(\text{eq})})}}$$

Every variant is therefore inflated above its base-population value $\frac{1}{2}\beta_k$ by the same factor $1/(1 - \rho_g^{(\text{eq})})$, whatever its effect size.

Justification for approximation: The neglected term, relative to the one it sits beside, is

$$\frac{2\mu^{(\text{eq})}\beta_k^2}{(1 - \rho_g^{(\text{eq})})^2} = \underbrace{\frac{2\rho_g^{(\text{eq})}}{1 - \rho_g^{(\text{eq})}}}_{\text{[Moderate-AM]}} \cdot \underbrace{\frac{\beta_k^2}{V_A^{(0)}}}_{\text{[Large-Me]}},$$

using $\mu^{(\text{eq})} = \rho_g^{(\text{eq})}/V_A^{(\text{eq})}$ together with the $V_A^{(\text{eq})} = V_A^{(0)}/(1 - \rho_g^{(\text{eq})})$ that this approximation itself yields below in (46). The first factor is bounded by [Moderate-AM] and the second is small by [Large-Me] via (41), so neither alone suffices. All that's required is that the product of the two is close to 0.

We'll make use of this approximation to obtain a clean formula for $V_A^{(\text{eq})}$, but first, we derive the exact formula. Expanding $\text{Var}[g_i]$ over all pairs of variants, every off-diagonal covariance is $4a_{kl}^{(\text{eq})}$ — that is (18) with $s_{kl}^{(\text{eq})} = a_{kl}^{(\text{eq})}$, the fixed point of (32) — while the diagonal is instead $1 + 2a_{kk}^{(\text{eq})}$ by (19). We apply the off-diagonal rule to every pair and then correct the one term it does not govern, using $a_{kl}^{(\text{eq})} = \mu^{(\text{eq})} c_k^{(\text{eq})} c_l^{(\text{eq})}$ from (31), $\mu^{(\text{eq})} V_A^{(\text{eq})} = \rho_g^{(\text{eq})}$ from (23), and $\sum_k \beta_k c_k^{(\text{eq})} = \frac{1}{2} V_A^{(\text{eq})}$ from (21):

$$\begin{aligned}
V_A^{(\text{eq})} &= \sum_k \sum_l \beta_k \beta_l \text{Cov}[x_{ik}, x_{il}] \\
&= \underbrace{\sum_k \sum_l \beta_k \beta_l 4a_{kl}^{(\text{eq})}}_{\text{one rule for every pair}} + \underbrace{\sum_k \beta_k^2 \left[(1 + 2a_{kk}^{(\text{eq})}) - 4a_{kk}^{(\text{eq})} \right]}_{\text{diagonal correction}} \\
&= 4\mu^{(\text{eq})} \left(\sum_k \beta_k c_k^{(\text{eq})} \right)^2 + \sum_k \beta_k^2 (1 - 2a_{kk}^{(\text{eq})}) \\
&= \underbrace{\mu^{(\text{eq})} (V_A^{(\text{eq})})^2}_{\rho_g^{(\text{eq})} V_A^{(\text{eq})}} + V_A^{(0)} - 2 \sum_k \beta_k^2 a_{kk}^{(\text{eq})} \\
V_A^{(\text{eq})} (1 - \rho_g^{(\text{eq})}) &= V_A^{(0)} - 2 \sum_k \beta_k^2 a_{kk}^{(\text{eq})}
\end{aligned} \tag{43}$$

$$\boxed{V_A^{(\text{eq})} = \frac{V_A^{(0)} - 2 \sum_k \beta_k^2 a_{kk}^{(\text{eq})}}{1 - \rho_g^{(\text{eq})}}}$$

We'll see in a later section that half of this numerator, $\frac{1}{2} (V_A^{(0)} - 2 \sum_k \beta_k^2 a_{kk}^{(\text{eq})})$, will come up again as the variance of the deviation of offspring genetic values from their parents' average. We use (31) to write a in terms of c and then plug in the approximated formula for c from (42):

$$\begin{aligned}
2 \sum_k \beta_k^2 a_{kk}^{(\text{eq})} &= 2 \mu^{(\text{eq})} \sum_k \beta_k^2 (c_k^{(\text{eq})})^2 \approx 2 \mu^{(\text{eq})} \sum_k \beta_k^2 \frac{\beta_k^2}{4(1 - \rho_g^{(\text{eq})})^2} \\
&= \frac{\mu^{(\text{eq})}}{2(1 - \rho_g^{(\text{eq})})^2} \sum_k \beta_k^4 \\
&= \frac{\rho_g^{(\text{eq})}}{2 V_A^{(0)} (1 - \rho_g^{(\text{eq})})} \sum_k \beta_k^4,
\end{aligned} \tag{44}$$

the last step using $\mu^{(\text{eq})} = \rho_g^{(\text{eq})} (1 - \rho_g^{(\text{eq})}) / V_A^{(0)}$, which is the exact $\mu^{(\text{eq})} = \rho_g^{(\text{eq})} / V_A^{(\text{eq})}$ with $V_A^{(\text{eq})}$ replaced by the very approximation we are in the middle of deriving. This is justified because this value of $V_A^{(\text{eq})}$ is used only to *measure the size* of a term we are about to discard, and the error in this discarded term is guaranteed to be smaller than the error in the term we are approximating.

Substituting that back into (43) and factoring out $V_A^{(0)}/(1 - \rho_g^{(\text{eq})})$,

$$V_A^{(\text{eq})} \approx \frac{V_A^{(0)}}{1 - \rho_g^{(\text{eq})}} \left[1 - \frac{\rho_g^{(\text{eq})}}{2(1 - \rho_g^{(\text{eq})})} \cdot \frac{\sum_k \beta_k^4}{(V_A^{(0)})^2} \right] \quad (45)$$

The last factor of (45) is exactly $1/M_e$ by (40), so the whole correction is $\rho_g^{(\text{eq})}/[2(1 - \rho_g^{(\text{eq})})M_e]$ — an error controlled by the effective number of variants rather than by any individual effect size.

[VA-inflation] The additive variance inflates by a single factor

The bracketed correction in (45) is negligible, leaving the additive variance at equilibrium as the base-population value scaled by one factor that depends on nothing but the genetic correlation between mates:

$$V_A^{(\text{eq})} = \frac{V_A^{(0)}}{1 - \rho_g^{(\text{eq})}} \quad (46)$$

Justification for approximation: Written with (40), the neglected term is a relative error in $V_A^{(\text{eq})}$ of

$$\frac{\rho_g^{(\text{eq})}}{2(1 - \rho_g^{(\text{eq})})} \cdot \frac{1}{M_e},$$

small by [Large-Me] and bounded by [Moderate-AM]. The error is signed: dropping the term *overstates* $V_A^{(\text{eq})}$, since a variant's own departure from Hardy–Weinberg drags its contribution down.

We can also write a formula for the equilibrium heritability:

$$h_{(\text{eq})}^2 = \frac{V_A^{(\text{eq})}}{V_P^{(\text{eq})}} = \frac{V_A^{(\text{eq})}}{V_A^{(\text{eq})} + V_E} = \frac{V_A^{(0)}/(1 - \rho_g^{(\text{eq})})}{V_A^{(0)}/(1 - \rho_g^{(\text{eq})}) + V_E}$$

$$h_{(\text{eq})}^2 = \frac{V_A^{(0)}}{V_A^{(0)} + (1 - \rho_g^{(\text{eq})})V_E} \quad (47)$$

$$h_{(\text{eq})}^2 = \frac{h_{(0)}^2}{h_{(0)}^2 + (1 - \rho_g^{(\text{eq})})(1 - h_{(0)}^2)}$$

$$h_{(\text{eq})}^2 = \frac{h_{(0)}^2}{1 - \rho_g^{(\text{eq})}(1 - h_{(0)}^2)}$$

These formulas for $V_A^{(\text{eq})}$ and $h_{(\text{eq})}^2$ are easily interpretable, but $\rho_g^{(\text{eq})}$ still relies on either $V_A^{(\text{eq})}$ or $h_{(\text{eq})}^2$ through (23). We can define $\rho_g^{(\text{eq})}$ purely in terms of the starting parameters to resolve the circularity.

$$\rho_g^{(\text{eq})} = \rho_y h_{(0)}^2 = \frac{\rho_y h_{(0)}^2}{1 - \rho_g^{(\text{eq})} (1 - h_{(0)}^2)} \quad (48)$$

$$\begin{aligned} \rho_g^{(\text{eq})} - (\rho_g^{(\text{eq})})^2 (1 - h_{(0)}^2) &= \rho_y h_{(0)}^2 \\ (1 - h_{(0)}^2) (\rho_g^{(\text{eq})})^2 - \rho_g^{(\text{eq})} + \rho_y h_{(0)}^2 &= 0 \end{aligned}$$

$$\boxed{\rho_g^{(\text{eq})} = \frac{1 - \sqrt{1 - 4 \rho_y h_{(0)}^2 (1 - h_{(0)}^2)}}{2(1 - h_{(0)}^2)}} \quad (49)$$

The $-$ root is the admissible one and the discriminant is never negative.

2.3 Covariance between relatives

Now that we have derived the statistical properties of the population at equilibrium under assortative mating, we will determine the expected covariance between pairs of individuals given some set of information, similar to Section 1 of this write-up.

2.3.1 Inheritance of genetic values

Rather than modelling the inheritance of allele states from parent to offspring, we will instead model the transmission of genetic values themselves. From here on we work only at equilibrium, and drop the (eq) superscript for legibility: V_A , V_P , h^2 , ρ_g and μ now denote the equilibrium values solved for above.

We can extend the logic of (13)–(15) to decompose the genetic value of an offspring as being the average of their parents' genetic values plus an independent residual term. Summing the transmitted alleles over variants, weighted by their effects, and substituting (15) for each one:

$$\begin{aligned} g_o &= \sum_k \beta_k x_{ok} = \sum_k \beta_k (z_{ok}^{(m)} + z_{ok}^{(p)}) \\ &= \sum_k \beta_k \left(\frac{1}{2} x_{mk} + \epsilon_{ok}^{(m)} + \frac{1}{2} x_{pk} + \epsilon_{ok}^{(p)} \right) \\ &= \frac{1}{2} g_m + \frac{1}{2} g_p + \underbrace{\sum_k \beta_k (\epsilon_{ok}^{(m)} + \epsilon_{ok}^{(p)})}_{\varepsilon_o} \end{aligned} \quad (50)$$

$$\boxed{g_o = \frac{1}{2} g_m + \frac{1}{2} g_p + \varepsilon_o}$$

Since each $\epsilon_{ok}^{(u)}$ is uncorrelated with *every* allele carried by the parent it came from, their weighted sum ε_o is also going to be uncorrelated with g_m and g_p and everything upstream

of it. As a consequence, we can easily compute the variance of this genetic value deviation, making use of (25):

$$\begin{aligned}\text{Var}[\varepsilon_o] &= \sum_k \sum_l \beta_k \beta_l \text{Cov}[\epsilon_{ok}^{(m)} + \epsilon_{ok}^{(p)}, \epsilon_{ol}^{(m)} + \epsilon_{ol}^{(p)}] \\ &= \sum_k \beta_k^2 \left(\text{Var}[\epsilon_{ok}^{(m)}] + \text{Var}[\epsilon_{ok}^{(p)}] \right) = \sum_k \beta_k^2 \left(\frac{1}{2} - a_{kk} \right)\end{aligned}\quad (51)$$

$$\boxed{\text{Var}[\varepsilon_o] = \frac{1}{2} V_A (1 - \rho_g) \approx \frac{1}{2} V_A^{(0)}}$$

Note that under **[VA-inflation]**, which is (46) rearranged to $V_A(1 - \rho_g) = V_A^{(0)}$, this is equivalent to half of the additive variance under random mating. This means that the residual genetic variance of a child around the parental mean is determined by the additive variance under random mating and therefore is not impacted by assortative mating.

Because ε_o is uncorrelated with both parents and across siblings, (50) is a linear model of exactly the kind Section 2.2.1 draws: two directed edges of $\frac{1}{2}$ into the offspring's genetic value, and a disturbance carrying (51). We can therefore draw a simplified path diagram that omits individual alleles and genotypes. We'll also omit the explicit environmental term and instead code it as exogenous variance on the phenotype.

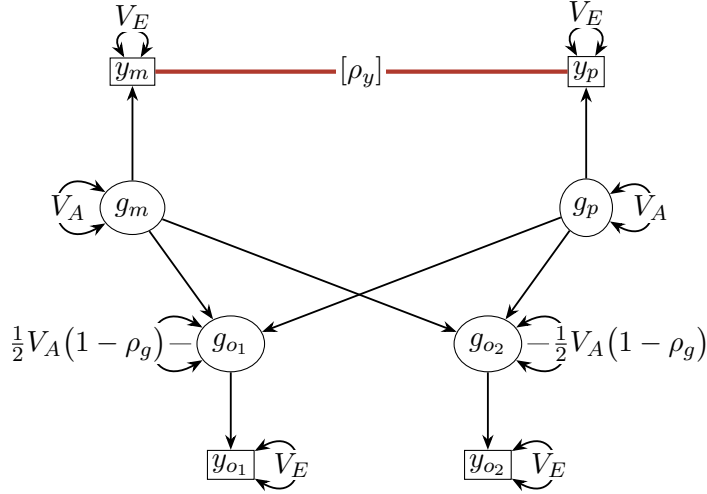


Figure 4: A mated pair and two of their offspring at equilibrium, with inheritance happening at the level of genetic values, rather than alleles. The environment appears only as the V_E on each phenotype's own self-loop, so $\text{Var}[y] = V_A + V_E$ as before.

Similar to Table 3, we can summarize the covariance between the genetic and phenotypic values across mate pairs and across parent–offspring pairs, now at equilibrium rather than in the first generation. These expressions are exact if we hold **[No-inbreeding]**, **[Discrete-generations]**, and **[AM-equilibrium]**.

	mate pair (m, p)	parent-offspring (m, o_1)
$\text{Cov}[g_m, g_*]$	$\rho_g V_A$	$\frac{1}{2} V_A (1 + \rho_g)$
$\text{Cov}[g_m, y_*]$	$\rho_y V_A$	$\frac{1}{2} V_A (1 + \rho_g)$
$\text{Cov}[y_m, g_*]$	$\rho_y V_A$	$\frac{1}{2} V_A (1 + \rho_y)$
$\text{Cov}[y_m, y_*]$	$\rho_y V_P$	$\frac{1}{2} V_A (1 + \rho_y)$

Table 5: Covariances read off Figure 4 at equilibrium, where * stands for the second member of the pair named by the column.

Note the asymmetry that arises: The covariance between a parent’s genetic value and their offspring’s phenotype is lower than the covariance between a parent’s phenotypic value and their offspring’s genetic value. This is because the parent’s environment (which is part of their phenotype) predicts the genetic value their offspring inherits from the other parent but not the offspring’s environmental value, which remains random under our current model.

To help us develop a generalizable framework later, we can also conceive of an abstract “mean-parent” that averages the genetic, environmental, and phenotypic values of a pair of mates:

$$g_{mp} = \frac{1}{2}(g_m + g_p), \quad e_{mp} = \frac{1}{2}(e_m + e_p), \quad y_{mp} = \frac{1}{2}(y_m + y_p) = g_{mp} + e_{mp} \quad (52)$$

We can obtain the variance of the mean-parent is obtainable from Table 5:

$$\begin{aligned} \text{Var}[g_{mp}] &= \frac{1}{4} \left(\text{Var}[g_m] + \text{Var}[g_p] + 2\text{Cov}[g_m, g_p] \right) = \frac{1}{4} (2V_A + 2\rho_g V_A) \\ \text{Var}[y_{mp}] &= \frac{1}{4} \left(\text{Var}[y_m] + \text{Var}[y_p] + 2\text{Cov}[y_m, y_p] \right) = \frac{1}{4} (2V_P + 2\rho_y V_P) \end{aligned} \quad (53)$$

$$\boxed{\text{Var}[g_{mp}] = \frac{1}{2} V_A (1 + \rho_g)}$$

$$\boxed{\text{Var}[y_{mp}] = \frac{1}{2} V_P (1 + \rho_y)}$$

We can then encode the genetic value of the offspring as being a linear function of this mean-parent’s genetic value, which collapses (50) to a single edge:

$$\boxed{g_o = g_{mp} + \varepsilon_o} \quad (54)$$

The residual ε_o still carries (51) and is still uncorrelated with both parents, hence with g_{mp} , so the two terms partition the offspring’s genetic variance exactly:

$$\text{Var}[g_o] = \text{Var}[g_{mp}] + \text{Var}[\varepsilon_o] = \frac{1}{2} V_A (1 + \rho_g) + \frac{1}{2} V_A (1 - \rho_g) = V_A \quad (55)$$

We can then similarly write a table for the covariance in genetic and phenotypic values of offspring to their own mean-parent.

	Var[·]	Cov[·, g_o]	Cov[·, y_o]
· = g_{mp}	$\frac{1}{2}V_A(1 + \rho_g)$	$\frac{1}{2}V_A(1 + \rho_g)$	$\frac{1}{2}V_A(1 + \rho_g)$
· = y_{mp}	$\frac{1}{2}V_P(1 + \rho_y)$	$\frac{1}{2}V_A(1 + \rho_y)$	$\frac{1}{2}V_A(1 + \rho_y)$

Table 6: Variances of a mean-parent and its covariances with one of its offspring, at equilibrium. The two right-hand columns reproduce the parent–offspring column of Table 5, since averaging two exchangeable parents returns the same covariance of each parent with the child they share.

2.3.2 Tracing paths between relatives

When tracing through a pedigree, three types of *steps* can be performed: going up a meiosis (offspring $\rightarrow$ parent), going down a meiosis (parent $\rightarrow$ offspring), and crossing a mating (mother — father). We will use these simple building blocks to construct a generalizable framework for expressing the covariance between any two individuals in a pedigree. First, we formalize the following notation and terminology.

Throughout, a parenthesized superscript names a generation and a bare superscript a number of steps.

- t_X denotes the generation that an individual X belongs to, according to [\[Discrete-generations\]](#).
- $m(X)$ and $p(X)$ denote the two parents, mother and father, of an individual X .
- $\mathcal{A}^{(t)}(X)$, defined for $t \leq t_X$, is the set of unique ancestors of X at generation t , which can be obtained recursively:

$$\mathcal{A}^{(t_X)}(X) = \{X\}, \quad \mathcal{A}^{(t-1)}(X) = \bigcup_{Y \in \mathcal{A}^{(t)}(X)} \{m(Y), p(Y)\}.$$

Dropping the superscript collects every generation, $\mathcal{A}(X) = \bigcup_{t \leq t_X} \mathcal{A}^{(t)}(X)$, so that $X \in \mathcal{A}(X)$.

- $\mu(X) = \{X\} \cup \{Y : X \text{ and } Y \text{ have mated}\}$ is the set of individuals that X has mated with, including themselves, so that μ is a single step of mating. Not every mating guarantees the existence of offspring, but every offspring guarantees the existence of a mating. Mates belong to the same generation, so a step of μ never leaves generation t_X .
- $\mathcal{M}(X)$ is the set of individuals that are either X or have mated with someone in $\mathcal{M}(X)$, so that $\mathcal{M}$ is arbitrarily many steps of mating. Writing μ^k for k steps, it can be defined recursively in the same shape as the ancestor sets:

$$\mu^0(X) = \{X\}, \quad \mu^k(X) = \bigcup_{Y \in \mu^{k-1}(X)} \mu(Y), \quad \mathcal{M}(X) = \bigcup_{k \geq 0} \mu^k(X).$$

The number of matings separating two individuals is the fewest steps that reach one from the other, $N_\mu(X, Y) = \min\{k : Y \in \mu^k(X)\}$, which is 0 when $X = Y$ and ∞ when $Y \notin \mathcal{M}(X)$.

- $\overline{\mathcal{M}}^{(t)}(X, Y)$ denotes all individuals of generation t which lie on a mating chain possessing an ancestor of X and a (possibly overlapping) ancestor of Y :

$$\overline{\mathcal{M}}^{(t)}(X, Y) = \left\{ Z : t_Z = t, \quad \mathcal{M}(Z) \cap \mathcal{A}^{(t)}(X) \neq \emptyset, \quad \mathcal{M}(Z) \cap \mathcal{A}^{(t)}(Y) \neq \emptyset \right\}.$$

Every member of a mating chain has the same $\mathcal{M}$, so this set is closed under mating: $\mu(Z) \subseteq \overline{\mathcal{M}}^{(t)}(X, Y)$ for each of its members Z .

In order for closed form solutions to be more manageable, we assume the following constraint to pedigree structure:

[Single-recent-mating-chain] Two individuals are connected by a single most recent mating chain

For any two probands A and B , we assume that there is some generation $t_{AB} \leq t_A, t_B$ for which:

- $\overline{\mathcal{M}}^{(t)}(A, B)$ is empty for $t > t_{AB}$, so that $t_{AB} = \max \{ t : \overline{\mathcal{M}}^{(t)}(A, B) \neq \emptyset \}$
- There are exactly two ancestors at generation t_{AB} of A which lie on a mating chain with an ancestor of B , and vice-versa. Using the short-hand $\widehat{\mathcal{M}}(A, B) = \overline{\mathcal{M}}^{(t_{AB})}(A, B)$,

$$\left| \mathcal{A}^{(t_{AB})}(A) \cap \widehat{\mathcal{M}}(A, B) \right| = \left| \mathcal{A}^{(t_{AB})}(B) \cap \widehat{\mathcal{M}}(A, B) \right| = 2$$

- The matings in $\widehat{\mathcal{M}}(A, B)$ do not form a loop. Each member Z sits at $|\mu(Z)| - 1$ matings and each mating is counted from both of its ends, so the chain is a tree exactly when its matings number one fewer than its members:

$$\frac{1}{2} \sum_{Z \in \widehat{\mathcal{M}}(A, B)} \left(|\mu(Z)| - 1 \right) = |\widehat{\mathcal{M}}(A, B)| - 1$$

Written this way the condition also forces the chain to be a single one, since a $\widehat{\mathcal{M}}(A, B)$ falling into two separate chains would have two fewer matings than members.

In more conversational terms, it means a single mating chain connects the two pairs of ancestors of each of two probands at the most recent generation possible. Furthermore, the mating chain must be linear and cannot form a loop.

Commonly bundled into: an outbred, loop-free pedigree.

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Suppose we have two distinct probands, A and B , of two possibly different generations, t_A and t_B . For either proband, we can look back one generation and identify the two parents of that proband, e.g. $m(A)$ and $p(A)$. We can repeat the same for each parent to obtain grandparents, and so on, identifying a maximum of $2^{(t-t_i)}$ unique ancestors at

generation t for each proband. Suppose that for some generation t_{AB} , the ancestors of A and B are such that at least one ancestor of A is either 1) also an ancestor of B , or 2) mated to an ancestor of B , or 3) mated to someone who mated an ancestor of B , and so on, for up to a theoretically infinite number of matings in a chain. So, for any pair of ancestor identified for each proband, A' and B' , we can quantify the number of matings N_μ that separate them, ranging from $N_\mu = 0$ when $A' = B'$ to $N_\mu = \infty$ when no chain of matings exist to connect A' and B' . In the below figure, we observe a family structure where $N_\mu = 0, 1, 2, 3$ can be observed:

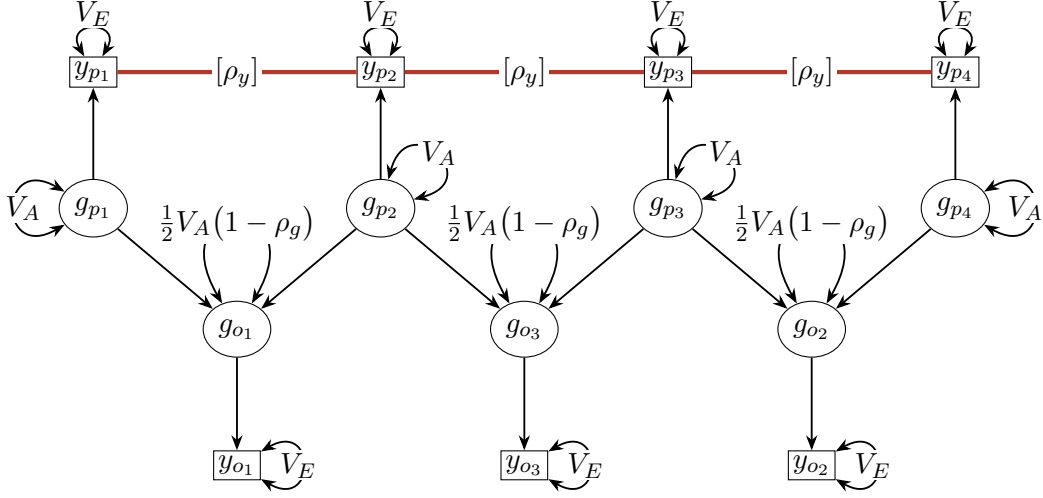


Figure 5: A pedigree of four parents joined by three matings, with one child per mating, drawn at the level of genetic values as in Figure 4. Reading left to right, p_1 and p_2 are the parents of o_1 , p_2 and p_3 of o_3 , and p_3 and p_4 of o_2 . Each parent is $N_\mu = 0$ to themselves. Each parent is $N_\mu = 1$ to their own mate, e.g. (p_1, p_2) ; $N_\mu = 2$ to their mate's other mate, e.g. (p_1, p_3) and (p_2, p_4) ; and $N_\mu = 3$ across the whole chain, (p_1, p_4) . In the offspring generation, by contrast, every pair is $N_\mu = \infty$: none of o_1, o_2, o_3 has mated, so no chain of matings connects any two of them.

Tracking four distances for every pair of probands is more bookkeeping than we need, so we summarise them by their average. For the two parent couples $\{m(A), p(A)\}$ and $\{m(B), p(B)\}$ there are four cross pairs, one ancestor drawn from each couple, and we call their average the *mean mating distance* between the couples:

$$\bar{N}_\mu = \frac{1}{4} \sum_{A' \in \{m(A), p(A)\}} \sum_{B' \in \{m(B), p(B)\}} N_\mu(A', B') \quad (56)$$

This average carries more information than an average usually would, because the four distances are not free to vary independently. Suppose the two couples share no individual and a single chain of matings joins them, and write k for the shortest of the four distances — the chain between the two nearest parents. The other three are then forced, since the only way to start or end elsewhere is to step across a couple's own mating: from the other parent of A one such step is needed, from the other parent of B likewise, and from both,

two. The four distances are therefore k , $k + 1$, $k + 1$ and $k + 2$, so that

$$\boxed{\bar{N}_\mu = k + 1} \quad (57)$$

Each additional mating separating the two couples raises $\bar{N}_\mu$ by exactly one. The only values below 2 belong to the two cases where the couples are not disjoint: two probands with the same parent couple have distances 0, 0, 1, 1 and so $\bar{N}_\mu = \frac{1}{2}$, and two probands sharing exactly one parent have 0, 1, 1, 2 and so $\bar{N}_\mu = 1$. A single number therefore names the configuration — $\frac{1}{2}$, 1, 2, 3, and so on — and we will use it as the index of the covariance below.

It is worth being clear about what $\bar{N}_\mu$ is and is not. It indexes the four distances rather than replacing them: the covariance derived below depends on $\sum \rho_y^{N_\mu}$ across the four cross pairs, and an average pins that sum down only because the four distances are locked together as above. Where they are not — where more than one chain of matings joins the same two individuals — the average and the covariance come apart, which is what [\[One-connection\]](#) below rules out.

For now, assume $t_A = t_B$ and that $t_{AB} = t_A - 1 = t_B - 1$; i.e., probands A and B are of the same generation and they each have at least one parent that has a chain of matings (including possibly $N_\mu = 0$) that connects them to at least one of the parents of the other proband. We can decompose the covariance between the phenotypes of A and B as being a product of each proband's covariance to their own mean-parent as well as the covariance of the two mean-parents to each other.

Because we have assumed independent environments ([\[Env-indep\]](#) and [\[No-IGE\]](#)), a proband's phenotypic covariance with anything upstream of it will be the same as the respective covariance to the proband's genetic value. Furthermore, we decomposed the proband's genetic value as the sum of their mean-parent genetic value plus an uncorrelated residual, meaning that this residual is also independent of factors upstream of it. Therefore, for proband A :

$$\boxed{\text{Cov}[y_A, Z] = \text{Cov}[g_A, Z] = \text{Cov}[g_{mp(A)}, Z]} \quad (58)$$

whenever $\text{Cov}[e_A, Z] = \text{Cov}[\varepsilon_A, Z] = 0$

Applying (58) at each proband in turn gives the decomposition:

$$\boxed{\text{Cov}[y_A, y_B] = \text{Cov}[g_{mp(A)}, g_{mp(B)}]} \quad (59)$$

Each proband's own factor has become one — by Table 6, a proband's covariance with its mean-parent, measured against that mean-parent's variance, is exactly unity at the genetic level — so the whole of the covariance now sits in the middle term.

It matters that the two ends land on g_{mp} rather than on y_{mp} . Written with mean-parent phenotypes, the two outer factors become the mid-parent regressions of Table 6, each equal to h^2 :

$$\text{Cov}[y_A, y_B] = h^2 \cdot \text{Cov}[y_{mp(A)}, y_{mp(B)}] \cdot h^2 \quad (60)$$

but this form holds only when the two mean-parents share no individual, which by (57) is exactly the condition $\bar{N}_\mu \geq 2$. Every route between two disjoint couples then leaves

through a mating, (12) splits it at the two matched phenotypes, and the factors of h^2 are exactly what converts those phenotypes back to genetic values. When $\bar{N}_\mu < 2$ the probands share a parent, that parent connects them through their genetic value directly with no mating to split at, and (60) understates the covariance.

Expanding (59) bilinearly leaves four terms, one for each pair of parents drawn one from each couple. Under the assumption above, every such pair sits in the same generation and is connected only by matings, so its covariance is obtained by crossing N_μ co-paths in a row: each crossing contributes a μ , each of the $N_\mu - 1$ individuals passed through contributes $\text{Var}[y] = V_P$, and each end contributes $\text{Var}[g] = V_A$.

$$\text{Cov}[g_{A'}, g_{B'}] = V_A \cdot \mu \cdot \underbrace{V_P \cdot \mu \cdots V_P \cdot \mu}_{N_\mu - 1 \text{ times}} \cdot V_A = \mu^{N_\mu} V_A^2 V_P^{N_\mu - 1} \quad (61)$$

$$\boxed{\text{Cov}[g_{A'}, g_{B'}] = \rho_g \rho_y^{N_\mu - 1} V_A}$$

The first crossing costs ρ_g and every further one costs ρ_y , because the connection is read at the genetic level only at its two ends; everyone passed through in between was matched on their whole phenotype.

Collecting the four terms, with A' ranging over $m(A)$ and $p(A)$ and B' over $m(B)$ and $p(B)$:

$$\boxed{\text{Cov}[y_A, y_B] = \frac{V_A}{4} \sum_{A'} \sum_{B'} f(N_\mu(A', B'))} \quad (62)$$

$$f(N_\mu) = \begin{cases} 1 & N_\mu = 0 \\ \rho_g \rho_y^{N_\mu - 1} & 1 \leq N_\mu < \infty \\ 0 & N_\mu = \infty \end{cases}$$

The three cases are the three things a cross pair of parents can be: the same person, for whom $\text{Cov}[g_{A'}, g_{A'}] = V_A$; two people joined by a chain of matings, giving (61); or two people with no such chain, who contribute nothing.

Because the four distances are locked together, (62) collapses to a single geometric law in $\bar{N}_\mu$. For two disjoint couples the distances are $k, k+1, k+1, k+2$, and substituting $k = \bar{N}_\mu - 1$:

$$\text{Cov}[y_A, y_B] = \frac{V_A}{4} (\rho_g \rho_y^{k-1} + 2\rho_g \rho_y^k + \rho_g \rho_y^{k+1}) = \frac{V_A}{4} \rho_g \rho_y^{k-1} (1 + \rho_y)^2 \quad (63)$$

$$\boxed{\text{Cov}[y_A, y_B] = \frac{V_A}{4} \rho_g \rho_y^{\bar{N}_\mu - 2} (1 + \rho_y)^2 \quad (\bar{N}_\mu \geq 2)}$$

The two factors of $1 + \rho_y$ are one per couple: at each end the chain either stops at the near parent or steps across that couple's own mating, and the choice is made independently at the two ends. Every further mating separating the couples then costs one power of ρ_y .

The two cases below $\bar{N}_\mu = 2$ do not follow this law, and the reason is worth naming. A cross pair at $N_\mu = 0$ contributes V_A , whereas (61) continued down to $N_\mu = 0$ would have contributed $h^2 V_A$; each shared parent therefore adds $(1 - h^2) V_A / 4$ beyond what the

mating law alone gives. That is the same discrepancy that made (60) conditional, and it has the same cause: a shared parent is a genetic identity, not a mating.

parents of A and B	four N_μ	$\bar{N}_\mu$	$\text{Cov}[y_A, y_B]$
the same couple	0, 0, 1, 1	$\frac{1}{2}$	$\frac{1}{2}V_A(1 + \rho_g)$
share one parent	0, 1, 1, 2	1	$\frac{1}{4}V_A(1 + 2\rho_g + \rho_g\rho_y)$
disjoint couples	$k, k+1, k+1, k+2$	$k + 1$	$\frac{1}{4}V_A \rho_g \rho_y^{\bar{N}_\mu - 2} (1 + \rho_y)^2$

Table 7: The covariance between two probands of the same generation, indexed by the mean mating distance $\bar{N}_\mu$ of (56). The three rows are the classical sibling families: full siblings are $\bar{N}_\mu = \frac{1}{2}$, half siblings 1, and step-siblings the first member of the third row at $\bar{N}_\mu = 2$ — the pairs (o_1, o_3) and (o_1, o_2) of Figure 5 instantiate the last two rows. Beyond that, each further mating separating the two couples adds one to $\bar{N}_\mu$ and one power of ρ_y . Under random mating $\rho_g = \rho_y = 0$ and the rows reduce to $\frac{1}{2}V_A$, $\frac{1}{4}V_A$ and 0: the third row exists only because of assortment, which is why no classical degree is defined for it.

Two conditions have been implicit in all of the above, and are worth stating now that we can say what each one buys. The first is what makes N_μ a function of the pedigree at all; the second says what to do when it is not.

[One-connection] Any pair of individuals is joined by a single route

Two individuals in a pedigree are connected by exactly one route through it. Ascending through a parent and ascending through that parent's mate count as one route and not two, so full siblings satisfy this; two couples cross-remarried to each other, or two siblings marrying two siblings, do not. In particular $N_\mu(A', B')$ is the length of the one chain of matings joining A' and B' , and the four distances entering (56) are then locked together as (57) describes.

Commonly bundled into: an outbred, loop-free pedigree; the [No-inbreeding] assumption; exclusion of doubled relationships.

[Shortest-chain] Only the shortest chain of matings is counted

Where more than one chain of matings joins two individuals, $N_\mu(A', B')$ is taken to be the length of the shortest of them and the rest are dropped.

Justification for approximation: Chains contribute additively, so two chains of lengths $k_1 \leq k_2$ give $\rho_g(\rho_y^{k_1-1} + \rho_y^{k_2-1})V_A$ in place of (61). Keeping only the shortest therefore understates the pair by the fraction $\rho_y^{k_2-k_1}/(1 + \rho_y^{k_2-k_1})$, which falls by a factor of about ρ_y for each unit by which the second chain is longer, and vanishes as assortment weakens. It reaches its worst case of $\frac{1}{2}$ only when two chains have exactly the same length, where the true covariance is double the one kept.

One case lies outside the reach of $\bar{N}_\mu$ altogether, and it is a matter of scope rather than of approximation. Assuming $t_{AB} = t_A - 1$ requires the two parent couples to be joined by matings. When they are joined by descent instead — first cousins, whose parents are siblings — no chain of matings connects any cross pair, every N_μ is infinite, and (62) returns zero, correctly reporting that nothing connects the two couples *within their own generation*. The remedy is not a different N_μ but a second application of (58), replacing each parent in turn by their own mean-parent.

2.3.3 Covariance implied by matings

Per our model of assortative mating ([Primary-AM]), matching acts at the level of the phenotype, so that a mated pair B and C satisfy

$$\boxed{\text{Corr}[y_B, y_C] = \rho_y} \quad (64)$$

and nothing else about them is matched directly.

As a consequence, suppose individuals A and D are only connected to each other through the mating of B and C . Then, by following Wright's path tracing rules — specifically (12), which says that crossing a co-path splits a covariance at the two matched phenotypes — we can decompose their covariance as

$$\boxed{\text{Cov}[y_A, y_D] = \text{Cov}[y_A, y_B] \cdot \mu \cdot \text{Cov}[y_C, y_D]} \quad (65)$$

where $\mu = \rho_y/V_P$ is the co-path coefficient of (11). The mating contributes one factor of μ , and everything on either side of it enters only through how it connects to the mated phenotype nearest it. This is what licenses splitting a path at its matings and treating the segments separately.

Furthermore, to obtain covariances involving one or both *genetic* values, we have to modify (65), because the legs then run to g rather than to y . The modification is confined to the ends: replacing y_A by g_A replaces the leg $\text{Cov}[y_A, y_B]$ by $\text{Cov}[g_A, y_B]$, and nothing in the interior changes. For the mated pair themselves this is immediate, since $\text{Cov}[g_B, y_B] = V_A$ by [GE-indep] while $\text{Var}[y_B] = V_P$:

$$\text{Cov}[y_B, y_C] = \mu V_P^2 = \rho_y V_P, \quad \text{Cov}[g_B, y_C] = \mu V_A V_P = \rho_y V_A, \quad (66)$$

$$\boxed{\text{Cov}[g_B, g_C] = \mu V_A^2 = \rho_g V_A}$$

so each end promoted from a genetic value to a phenotype trades a factor of V_A for V_P — equivalently, promotes one ρ_g to ρ_y . We will meet this trade again at the ends of a whole path, where it is the only place the distinction between g and y survives.

2.3.4 Covariance implied by segments

We begin by deriving the covariance between genetic values of two individuals, A and B , at the ends of a segment. Trivially, if A and B are the same individual, then $\text{Cov}[g_A, g_B] = V_A$ (and $\text{Cov}[y_A, y_B] = V_P$). Now consider that B is the parent of

A. Per our path tracing rules, their genetic values covary both through the directed edge $B \rightarrow A$ and through A's other parent C , who under assortative mating is themselves genetically similar to B . Writing A's genetic value with (50):

$$\begin{aligned} \text{Cov}[g_A, g_B] &= \text{Cov}\left[\frac{1}{2}g_B + \frac{1}{2}g_C + \varepsilon_A, g_B\right] \\ &= \frac{1}{2}\underbrace{\text{Var}[g_B]}_{V_A} + \frac{1}{2}\underbrace{\text{Cov}[g_C, g_B]}_{\rho_g V_A} + \underbrace{\text{Cov}[\varepsilon_A, g_B]}_0 \end{aligned} \quad (67)$$

$$\text{Cov}[g_A, g_B] = \frac{1 + \rho_g}{2} V_A$$

The middle term is the whole effect of assortment here: were B 's mate chosen at random, $\text{Cov}[g_C, g_B]$ would vanish and we would recover the familiar halving.

Note that to obtain this formula for a segment, we had to call upon a mating already. This is not a violation of the definition but an instance of it: ascending to B and ascending to B 's mate C are two traversals of the same path, so the mating between them is consolidated into the single factor $\frac{1}{2}(1 + \rho_g)$ that describes one step of descent.

We can then extend this logic to any pair of *lineal* relatives — one descended from the other, so that the segment between them runs only upward or only downward. We multiply (67) once for each of the N_m meioses needed to connect them.

$$\text{Cov}[g_A, g_B] = V_A \cdot \underbrace{\frac{1 + \rho_g}{2} \cdot \frac{1 + \rho_g}{2} \cdots \frac{1 + \rho_g}{2}}_{N_m \text{ meioses}} \quad (68)$$

$$\text{Cov}[g_A, g_B] = \left(\frac{1 + \rho_g}{2}\right)^{N_m} V_A \quad (A, B \text{ lineal})$$

For *collateral* relatives — neither descended from the other, but still sharing a common ancestor — we cannot chain the up and down meioses without first asking whether the segment is *full*, turning at a couple both of whom are common ancestors, or *half*, turning at a single shared ancestor. A full segment appears at first to break the definition, since two routes connect its ends, one through each member of the couple; by the equivalence of Section ?? these are one path, and the segment records the fact by being full.

It is tempting to fold this into the degree d by declaring a full segment worth one degree less than a half one, and under random mating that is exactly right: a full segment carries precisely twice what a half segment of the same N_m carries, which is what 2^{-d} demands. Under assortment it is not. Assortment inflates a half segment more than a full one — the two *unshared* parents of a half pair are themselves mates of the shared parent, and so are correlated — and at $h^2 = \rho_y = \frac{1}{2}$ the ratio is nearer 1.5 than 2. A single exponent cannot absorb that, so we do not ask it to.

Instead we keep one exponent, N_m , and put the difference between the three kinds of

turn into a separate factor K :

$$\boxed{\text{Cov}[g_A, g_B] = \left(\frac{1 + \rho_g}{2}\right)^{N_m} K V_A} \quad (69)$$

$$K = \begin{cases} 1, & \text{the turn is one of the segment's own ends (lineal)} \\ \frac{2}{1 + \rho_g}, & \text{the turn is a couple (full)} \\ \frac{1 + 2\rho_g + \rho_y\rho_g}{(1 + \rho_g)^2}, & \text{the turn is a single ancestor (half)} \end{cases} \quad (70)$$

Equations (69)–(70) are exact for all three kinds of turn, which the degree formulation is not.

The degree has not been lost; it has been located. As $\rho_g \rightarrow 0$ the three cases of (70) become 1, 2 and 1, so under random mating K is a power of two and can be absorbed into the exponent — a full segment simply counts as one meiosis fewer. That is the classical d , and (69) reduces to $2^{-d}V_A$. Assortment is what pulls K off the powers of two, and once it does, the honest thing is to carry it separately rather than to round it back into an integer. The half case is Nagylaki's multiplier [10], also derived by path tracing in the supplement to Sunde et al. [8]; we meet it again in Section 2.3.7.

Since $\rho_y \geq \rho_g$, the half factor is at least one: half relatives are always *more* similar than their degree alone would predict. It is also the first quantity in this section that cannot be written in ρ_g alone — ρ_y appears on its own, and the two are proportional only through h^2 .

The following assumption has already been implicit in prior sections, but we state it explicitly here.

[Sex-exchangeable] Maternal and paternal connections are interchangeable

When tracing a path between relatives, the sex of the parent(s) that connect them does not matter.

Commonly bundled into: no maternal effects, no sex-specific assortment, and no sex difference in effect size.

2.3.5 Assembling a whole path

We now have both halves: (65) splits a path at its matings, and (69) evaluates each of the resulting segments. Putting them together, a path with n matings and $n + 1$ segments carries

$$\boxed{\text{Cov}[g_A, g_B] = \mu^n \prod_{i=0}^n S_i} \quad (71)$$

where S_i is the covariance carried by segment i . The μ^n is one factor per mating and the product is one factor per segment, so a path is read off left to right with nothing left over.

Two adjustments distinguish S_i from the bare (69), and both concern the ends of a segment where it meets a mating.

The first is that an interior segment consisting of a *single individual* contributes that individual's whole phenotypic variance:

$$S_i = V_P \quad (N_m = 0, \text{ the path passes through this person}). \quad (72)$$

The path arrives at their phenotype and leaves from it, so both legs attach to y rather than to g . This is the case that makes two people who share a mate resemble each other at all.

The second is that a meiosis *adjacent to a mating* may cost $\frac{1}{2}(1+\rho_y)$ rather than $\frac{1}{2}(1+\rho_g)$. It does so when the individual at that end of the segment is a *parent* on the segment — that is, when the path leaves them through one of their children. The reason is the same one that gave (66) its V_P : a mate of that person was matched to their whole phenotype, environment included, and the child on the path carries that mate's genes. When instead the individual at the mating end is a *child* on the segment, their own environment plays no part and the factor stays $\frac{1}{2}(1+\rho_g)$. Writing $p_i \in \{0, 1, 2\}$ for the number of a segment's mating-facing ends at which the individual is a parent,

$$S_i = V_A K_i \left(\frac{1+\rho_y}{2} \right)^{p_i} \left(\frac{1+\rho_g}{2} \right)^{N_m^{(i)} - p_i} \quad (73)$$

with K_i from (70). A path with no matings has $n = 0$, one segment, $p = 0$, and (73) collapses back to (69).

2.3.6 Genetic values, phenotypes, and the ends of a path

Everything so far has been a covariance between genetic values. The quantity we usually want is the phenotypic correlation, and the step between them is not always a division by V_P .

The rule is local to the two outer ends of the path, and it asks one question of each: *is that individual's own mating used by the path?*

If it is not — if at least one meiosis separates them from their segment's turn and from any mating — then their environment is uncorrelated with everything the path reaches, so

$$\text{Cov}[y_A, y_B] = \text{Cov}[g_A, g_B], \quad \text{Corr}[y_A, y_B] = \frac{\text{Cov}[g_A, g_B]}{V_P} = h^2 \frac{\text{Cov}[g_A, g_B]}{V_A}.$$

This covers every collateral pair and every in-law pair: full siblings, cousins, half siblings, step-siblings.

If it is — if the individual sits directly in a mating the path uses — then their environment does reach the other end, and promoting g to y at that end promotes one ρ_g to ρ_y . Two

cases arise in practice. For a *mated pair* both ends are of this kind, and (66) already gave the answer. For a *lineal* pair the ancestor is of this kind and the descendant is not, so exactly one factor changes, however many generations separate them:

$$\text{Corr}[y_A, y_B] = h^2 \left(\frac{1 + \rho_g}{2} \right)^{N_m - 1} \left(\frac{1 + \rho_y}{2} \right) \quad (A, B \text{ lineal}) \quad (74)$$

The *genetic* covariance of a lineal pair is still (69) with $K = 1$; it is only the phenotypic correlation that departs. Since $\rho_y \geq \rho_g$, a lineal pair is always more similar than a collateral pair with the same N_m — a grandparent resembles a grandchild more than an aunt resembles a niece.

2.3.7 Common relative classes

Table 8 collects what (71)–(74) give for the pairs one actually meets, and Figure 5 shows a single pedigree containing most of them.

class	n	turn(s)	$\text{Corr}[y_A, y_B]$
mated pair	1	—	ρ_y
lineal	0	an end	$h^2 \left(\frac{1 + \rho_g}{2} \right)^{N_m - 1} \left(\frac{1 + \rho_y}{2} \right)$
collateral, full	0	a couple	$h^2 \left(\frac{1 + \rho_g}{2} \right)^{N_m} \frac{2}{1 + \rho_g}$
collateral, half	0	one ancestor	$h^2 \left(\frac{1 + \rho_g}{2} \right)^{N_m} \frac{1 + 2\rho_g + \rho_y \rho_g}{(1 + \rho_g)^2}$
in-law	1	two ends	$h^2 \rho_g \left(\frac{1 + \rho_y}{2} \right)^2 \left(\frac{1 + \rho_g}{2} \right)^{N_m - 2}$

Table 8: Expected phenotypic correlation by class, at equilibrium, for a pair whose only connection is the one named. Every row is (71) evaluated on the relevant path, and the classes differ only in the turn factor K and in how many meioses sit next to a mating and so carry ρ_y in place of ρ_g : none for a full collateral pair, one for a lineal pair, two for an in-law pair. Under random mating $\rho_g = \rho_y = 0$ and every row with a common ancestor collapses to $h^2 2^{-d}$, with d as in the classical definition; the in-law row collapses to zero.

Two remarks close the section.

The first is that N_m is doing the work a *modified degree* is sometimes introduced to do. In-law pairs have no common ancestor, so the classical d is undefined for them; Eftedal et al. [11] therefore index them by the combined distance to the connecting couple. That quantity is exactly N_m , which we already had, so no new index is needed — and N_m is defined for every pair, with or without a common ancestor.

The second is that the in-law row disagrees with the expression Eftedal et al. [11] fit, which is $h^2 \rho_g \left(\frac{1}{2}(1 + \rho_g) \right)^{N_m}$ — the ordinary rate factor raised to the whole distance, with

no ρ_y anywhere. The difference is the two meioses adjacent to the connecting mating, and it makes our expression the larger: at $h^2 = \rho_y = \frac{1}{2}$, a step-sibling correlation of 0.070 against their 0.049. The disagreement is not an algebraic slip on their part but a modelling one, and it is internal to their own treatment: they import Nagylaki's half-sibling multiplier, which carries the phenotypic mate correlation as a parameter distinct from the genotypic one, and then derive in-laws as though the two were the same. Under assortment on the phenotype they are not, and the same correction belongs in both places.

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2.4 Level 0 — unchanged

Conditioning on both individuals' causal genotypes and the true effect sizes leaves nothing to average over, so (4) holds exactly as written. Level 0 is a statement about the genotype-to-phenotype map alone, and no mating rule can disturb it. That it survives every architecture untouched is a feature of where the tower starts, not a coincidence.

2.5 Level 1 — the off-diagonal no longer collapses

Section 1 moved from Level 0 to Level 1 by averaging over the effects and observing that the off-diagonal $k \neq k'$ terms vanish. Two separate clauses of [\[Random-effects\]](#) were needed for that: the effects are marginally uncorrelated, $E[\beta_k \beta_{k'}] = 0$, and their cross-products are independent of the genotypes, $E[\beta_k \beta_{k'} \mid \mathbf{X}] = E[\beta_k \beta_{k'}]$.

The first clause survives assortative mating. Assortment does not reach back and change the effect sizes; it changes which genotypes are common.

The second clause does not survive, and it is worth seeing why the failure is unavoidable rather than incidental. Under assortment, the correlations that build up between variants are generated *by* the effects: it is precisely the trait-increasing alleles that end up travelling together. The observed pattern of correlation in $\mathbf{X}$ is therefore informative about which variants have effects and with what signs, so conditioning on $\mathbf{X}$ shifts our belief about $\beta_k \beta_{k'}$. Formally, $E[\beta_k \beta_{k'} \mid \mathbf{X}]$ is positive for pairs of variants that are positively correlated in the sample, and the off-diagonal sum no longer disappears:

$$E[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] = \underbrace{\sum_k E[\beta_k^2] x_{ik} x_{jk}}_{V_A^{(0)} A_{ij}} + \underbrace{\sum_k \sum_{k' \neq k} E[\beta_k \beta_{k'} \mid \mathbf{X}] x_{ik} x_{jk'}}_{\text{disequilibrium term}}. \quad (75)$$

The first term is the Level-1 result of Section 1, with the additive variance replaced by the *genic* variance $V_A^{(0)}$ of (??) — the genotype matrix on its own can only recover the per-variant contributions. The second term is new, and it is what the remainder of the disequilibrium D lives in.

Note the shape of (75): two relatedness-like quantities added together, not one quantity rescaled. This is the form that methods for unrelated individuals take, where the second term is estimated with a second relatedness matrix alongside the GRM [12]. We do not develop it here.

Two things to settle before writing out the disequilibrium term explicitly. First, the natural pair-level definition weights $x_{ik}x_{jk'}$ by the population LD between k and k' ; the matrix $\mathbf{G}^2$ used in practice is an estimator of this that sums over *other* individuals, and so is a property of the sample rather than of the pair — which sits awkwardly with this article’s framing. Second, it needs a decision about whether to restrict to between-chromosome pairs, as is done in practice to avoid confounding with LD- and MAF-dependent architecture.

2.6 The tower re-orders

At this point Section 1 descended to realized relatedness and then to genetic degree. That route is no longer available, and the reason is the screening property named at the end of Section 1.

Under random mating, π_{ij} screened off d_{ij} : once we knew what fraction of the genome a pair actually shared, their pedigree told us nothing more. Under assortative mating this is false. The disequilibrium contribution to a pair’s covariance is carried by the correlations their *ancestors* had, and how many rounds of assortment separate them — information that lives in the pedigree and is not recoverable from how much genome they happen to share. Two pairs with identical π_{ij} but different pedigrees have different expected covariances.

The four information sets of Section 1 are therefore no longer a chain. Level 2 (realized relatedness) and Level 3 (pedigree) are now two *parallel* coarsenings of a common refinement — knowing both — and neither is a function of the other. We keep the level numbering, since each level still refers to the same information, but the order of derivation reverses: the pedigree is now the easy case and must be done first, the joint case follows by refinement, and realized relatedness alone is the hard case and comes last.

2.7 Level 3 — pedigree

One more generation

Everything at this level follows from a single recursion.

Lemma. *Let C be the child of P and P' , and let X be any random variable such that (i) $\text{Cov}[X, e_P] = 0$, and (ii) all association between X and P' runs through P ’s phenotype. Then*

$$\text{Cov}[X, g_C] = \frac{1 + \rho_g}{2} \text{Cov}[X, g_P]. \quad (76)$$

Proof. By (??), $\text{Cov}[X, g_C] = \frac{1}{2}\text{Cov}[X, g_P] + \frac{1}{2}\text{Cov}[X, g_{P'}]$, the segregation term dropping out. For the second piece, condition on y_P : by (ii), $\text{Cov}[X, g_{P'}] = \text{Cov}[X, \text{E}[g_{P'} \mid y_P]]$,

and the regression of a partner's genetic value on one's own phenotype has coefficient $\text{Cov}[y_P, g_{P'}]/V_P^{(\text{eq})} = \rho_y V_A^{(\text{eq})}/V_P^{(\text{eq})} = \rho_g$ by (??). Hence $\text{Cov}[X, g_{P'}] = \rho_g \text{Cov}[X, y_P] = \rho_g \text{Cov}[X, g_P]$, the last step using (i). $\square$

Condition (ii) is a restriction on the pedigree, and it has a familiar name.

[Monogamy] Each individual has children with one partner

Any two relatives are connected by a single line of descent, so a chain of association between them traverses at most one assortment link that is not on that line.

Commonly bundled into: a pedigree with no half-sibships; the leading real-world violation is an individual having children with more than one partner.

Collateral relatives

For two *collateral* relatives — individuals related through a common ancestor, neither descended from the other — Section ?? showed that no environmental cross terms arise, so (3) still applies and we need only the genetic covariance. The base case is a pair of full siblings, children of m and f . Expanding both siblings with (??),

$$\begin{aligned} \text{Cov}[g_i, g_j] &= \frac{1}{4} \left(\text{Var}[g_m] + \text{Var}[g_f] + 2\text{Cov}[g_m, g_f] \right) \\ &= \frac{1}{4} \left(V_A^{(\text{eq})} + V_A^{(\text{eq})} + 2\rho_g V_A^{(\text{eq})} \right) = V_A^{(\text{eq})} \frac{1 + \rho_g}{2}, \end{aligned}$$

using that the two siblings' transmissions from a given parent are conditionally independent given that parent, exactly as in Section 1. Applying (76) once for each further generation of separation gives, for a collateral pair of degree d_{ij} ,

$$\boxed{\text{E}[y_i y_j \mid d_{ij}] = V_A^{(\text{eq})} \left(\frac{1 + \rho_g}{2} \right)^{d_{ij}}} \quad (77)$$

in agreement with Sunde et al. [8]. Setting $\rho_g = 0$ recovers $V_A^{(0)} 2^{-d_{ij}}$, the Level-3 result of Section 1.

Lineal relatives

For a pair in which a is an ancestor of b , the environmental cross term of Section ?? contributes and the base case changes. For a parent and their offspring it is easiest to work with $\text{Cov}[y_m, g_o]$ directly, since e_o is a fresh draw:

$$\begin{aligned} \text{Cov}[y_m, y_o] &= \text{Cov}[y_m, g_o] = \frac{1}{2} \left(\text{Cov}[y_m, g_m] + \text{Cov}[y_m, g_f] \right) \\ &= \frac{1}{2} \left(V_A^{(\text{eq})} + \rho_y V_A^{(\text{eq})} \right) = V_A^{(\text{eq})} \frac{1 + \rho_y}{2}. \end{aligned}$$

The second term is the whole story: the focal parent's *phenotype* predicts the other parent's genetic value with regression coefficient $\rho_y V_A^{(\text{eq})}/V_P^{(\text{eq})}$, and half of those genes reach the child. Iterating (76) for the remaining generations gives

$$\boxed{\mathbb{E}[y_a y_b \mid d_{ab}] = V_A^{(\text{eq})} \left(\frac{1 + \rho_y}{2} \right) \left(\frac{1 + \rho_g}{2} \right)^{d_{ab}-1}} \quad (78)$$

Comparing (77) and (78), the two differ in one factor, ρ_y in place of ρ_g . Since $\rho_g = \rho_y h_{(\text{eq})}^2 < \rho_y$, the lineal covariance is always the larger. This is a genuine and somewhat surprising consequence: under random mating a parent–offspring pair and a full-sibling pair have identical expected covariance $V_A/2$, whereas under assortative mating the parent–offspring covariance exceeds the sibling covariance by

$$V_A^{(\text{eq})} \frac{\rho_y - \rho_g}{2} = \frac{\rho_y V_A^{(\text{eq})} (1 - h_{(\text{eq})}^2)}{2}.$$

In the worked example this is a 15% excess over the sibling value. The asymmetry is purely environmental in origin — it is the parent's own environment, correlated with the other parent's genes — and a model that tracked only genetic covariances would miss it entirely.

Degree is no longer sufficient

Equations (77) and (78) disagree at the same d , which already shows that the degree alone does not determine the answer. Half-siblings make the point again, from the other direction. They are second-degree relatives, but they violate [\[Monogamy\]](#), and the two outer parents P_1 and P_2 — both of whom assorted on the shared parent P — are themselves correlated:

$$\text{Cov}[g_{P_1}, g_{P_2}] = \text{Var}[y_P] \left(\frac{\rho_y V_A^{(\text{eq})}}{V_P^{(\text{eq})}} \right)^2 = \rho_g^2 V_P^{(\text{eq})},$$

which gives

$$\mathbb{E}[y_i y_j \mid \text{half-sibs}] = \frac{1}{4} \left(V_A^{(\text{eq})} (1 + 2\rho_g) + \rho_g^2 V_P^{(\text{eq})} \right).$$

This exceeds the collateral value $\frac{1}{4} V_A^{(\text{eq})} (1 + \rho_g)^2$ by $\frac{1}{4} \rho_g^2 V_E$. So three different second-degree relationships — grandparent–grandchild, avuncular, and half-sibling — have three different expected covariances.

[\[Genetic-degree\]](#) must therefore be strengthened: the conditioning variable at Level 3 is not the degree but the **pedigree relationship**, of which the degree is only a summary. The same calculation also shows that individuals with no common ancestor at all can be correlated: two people who both had children with the same person have expected phenotypic cross-product $\rho_y^2 V_P^{(\text{eq})}$, and partners themselves $\rho_y V_P^{(\text{eq})}$. Under random mating both are zero.

2.8 Levels 2 and 3 together — pedigree and realized relatedness

Knowing the pedigree fixes the *expected* sharing, $\bar{\pi}_d = 2^{-d}$, but two pairs with the same relationship still differ in how much genome they actually share. Section 1 handled that variation at Level 2; here we ask what it buys once the pedigree is already known. Following the natural parameterisation, we write the realized sharing as its pedigree expectation plus a deviation,

$$\pi_{ij} = \bar{\pi}_{d_{ij}} + \delta_{ij}, \quad \mathbb{E}[\delta_{ij} \mid d_{ij}] = 0.$$

The two terms of (75) respond to δ_{ij} very differently, and that is the whole content of this level. The genic term is a sum of *same-variant* products, and by [Same-locus-AM] the argument of Section 1 goes through unchanged: its conditional expectation is $V_A^{(0)}\pi_{ij}$, tracking the realized sharing exactly. The disequilibrium term is a sum of *cross-variant* products, and it behaves quite differently.

[GPD-pedigree] The disequilibrium contribution depends only on the pedigree

The cross-variant part of a pair’s covariance is a function of their pedigree relationship alone, and does not depend on the realized sharing π_{ij} .

Justification for approximation: A cross-variant term $\mathbb{E}[x_{ik}x_{jk'}]$ with $k \neq k'$ is determined by *which* ancestral genomes the two alleles descend from — fixed by the pedigree — and not by whether i and j happen to be IBD at either locus, since the two alleles are at different loci. A residual dependence does exist, because a pair sharing more genome is also more likely to be IBD at k and k' jointly, but for unlinked variants this is a second-order effect: the correction is of order $\rho_g\delta_{ij}$, being the product of the disequilibrium share (??) and the sharing deviation. Deriving it exactly is left open.

Putting the two together, the expected cross-product is the pedigree value plus a term linear in the sharing deviation. The coefficient on that term is fixed by requiring consistency with Level 3: averaging over δ_{ij} must return the Level-3 value for that relationship. Hence, writing $C_{ij}(\text{pedigree})$ for the Level-3 value of the pair’s relationship,

$$\mathbb{E}[y_i y_j \mid \text{pedigree}, \pi_{ij}] = \underbrace{C_{ij}(\text{pedigree})}_{\text{from (77) or (78)}} + V_A^{(0)} (\pi_{ij} - \bar{\pi}_{d_{ij}}) \quad (79)$$

The structure of (79) is worth dwelling on, because it separates two things that Section 1 could not tell apart. The *baseline* is set by the pedigree and is inflated by assortment. The *response to realized sharing* is not inflated at all: its coefficient is $V_A^{(0)}$, the base-population — equivalently genic — additive variance, because only the same-variant part of the covariance responds to how much genome a pair actually shares. Under random mating the two coincide, $V_A^{(0)} = V_A^{(\text{eq})}$, and (79) collapses back to $V_A\pi_{ij}$.

This suggests a way to see assortative mating directly, which we note without pursuing. Within a single relationship class, regressing pairs’ phenotypic cross-products on their

realized relatedness gives a slope that estimates $V_A^{(0)}$ and an intercept that carries the assortment-inflated baseline. The two would agree under random mating, so the gap between them is a signature of assortment, obtainable without ever comparing across relationship classes. Whether the residual $O(\rho_g \delta_{ij})$ term of **[GPD-pedigree]** is small enough for this to be clean is the open question flagged in that box.

2.9 Level 2 — realized relatedness alone

The last case is the one Section 1 found easiest and assortative mating makes hardest: we observe how much genome a pair shares, but know nothing about their pedigree. Because the pedigree is exactly what carries the disequilibrium contribution, we cannot evaluate (79) directly; we have to infer a pedigree from the sharing.

The natural device is to invert the degree-to-relatedness map of Section 1. Since $\bar{\pi}_d = 2^{-d}$, a pair sharing a fraction π_{ij} of their genome is, on average, as related as a relative of degree $-\log_2 \pi_{ij}$.

[Effective-degree] Realized sharing stands in for a degree

A pair's realized relatedness is treated as though it arose from a collateral relationship of *effective degree*

$$d_{ij}^{\text{eff}} := -\log_2 \pi_{ij},$$

which is generally not an integer.

Justification for approximation: The substitution is exact whenever π_{ij} equals its pedigree expectation for some collateral relationship, i.e. on the grid $\pi \in \{2^{-1}, 2^{-2}, \dots\}$; off the grid it interpolates. The interpolation is benign for distant pairs, where many relationships of similar degree contribute and the sharing distribution is smooth, and poor for close pairs, where a handful of discrete relationships dominate and, as §2.7 showed, relationships of the same degree can differ. It is therefore an approximation for the distant tail, not a replacement for (79).

Substituting d_{ij}^{eff} into (77) and simplifying,

$$\left(\frac{1 + \rho_g}{2}\right)^{-\log_2 \pi_{ij}} = 2^{\log_2 \pi_{ij}} \cdot (1 + \rho_g)^{-\log_2 \pi_{ij}} = \pi_{ij} \cdot \pi_{ij}^{-\log_2(1 + \rho_g)},$$

so that the expected cross-product is a *power law* in the realized relatedness:

$$\boxed{\mathbb{E}[y_i y_j \mid \pi_{ij}] = V_A^{(\text{eq})} \pi_{ij}^{1 - \log_2(1 + \rho_g)}} \quad (80)$$

Three things are worth reading off (80).

First, it contains Section 1 as a special case. With $\rho_g = 0$ the exponent is exactly 1, the relationship is linear, and we recover $\mathbb{E}[y_i y_j \mid \pi_{ij}] = V_A \pi_{ij}$. Assortative mating shows up as a single number: a deficit in the exponent.

Second, the exponent is less than one, so the curve is concave. The proportional inflation grows as pairs become more distant — in the worked example the exponent is 0.823, and while sibling covariance is inflated by 30% relative to a randomly mating population with the same base heritability, first-cousin covariance is inflated by 66%. This is the reason assortative mating is more visible in distant relatives than in close ones, and the reason it inflates estimates that lean on distant or nominally unrelated pairs.

Third, (80) is linear on a log–log scale,

$$\log E[y_i y_j \mid \pi_{ij}] = \log V_A^{(\text{eq})} + (1 - \log_2(1 + \rho_g)) \log \pi_{ij},$$

so the slope of expected cross-product against realized relatedness, both on log scales, estimates ρ_g through its shortfall below one. Combined with the within-class result of §2.8, this gives two distinguishable signatures: *within* a relationship class the response to sharing is linear with slope $V_A^{(0)}$, while *across* classes it is a power law with exponent below one. Under random mating these are the same line. The wedge between them is assortative mating.

Finally, note where (80) points as $\pi_{ij} \rightarrow 0$. The covariance vanishes, but more slowly than linearly, and the pairs contributing in that limit are the nominally unrelated ones. This is the regime in which the additive form (75) is the natural description, and where methods that fit a second, disequilibrium relatedness matrix alongside the GRM operate [12].

2.10 Summary

The same four information sets as Section 1, but no longer arranged in a chain. Level 2 and Level 3 are parallel coarsenings of their common refinement, and the derivation runs $0 \rightarrow 1 \rightarrow 3 \rightarrow (2, 3) \rightarrow 2$ rather than top to bottom.

Level	Conditioned on	$E[y_i y_j \mid \cdot]$	New assumptions
0	causal genotypes, effects	$g_i g_j$	— (unchanged)
1	genotypes $\mathbf{x}_i, \mathbf{x}_j$	$V_A^{(0)} A_{ij} + \text{disequilibrium}$	[Random-effects] weakened
3	pedigree	$V_A^{(\text{eq})} \left(\frac{1+\rho_g}{2}\right)^d$ (collateral)	[Pheno-AM], [Primary-AM], [AM-equilibrium], [Monogamy]
(2, 3)	pedigree and π_{ij}	$C_{ij} + V_A^{(0)}(\pi_{ij} - \bar{\pi}_d)$	[Same-locus-AM], [GPD-pedigree]
2	π_{ij}	$V_A^{(\text{eq})} \pi_{ij}^{1-\log_2(1+\rho_g)}$	[Effective-degree]

Reduced to one sentence: assortative mating adds a second channel of resemblance that travels through the pedigree rather than through shared segments, so relatedness and genealogy stop being interchangeable. Everything else in this section is bookkeeping for that one fact.

Still to do in this section: the explicit disequilibrium relatedness matrix at Level 1 and its estimation in samples of unrelated individuals; the exact size of the residual term in **[GPD-pedigree]**; and the non-equilibrium case, where a mismatch between observed and predicted correlations dates the onset of assortment [8].

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